

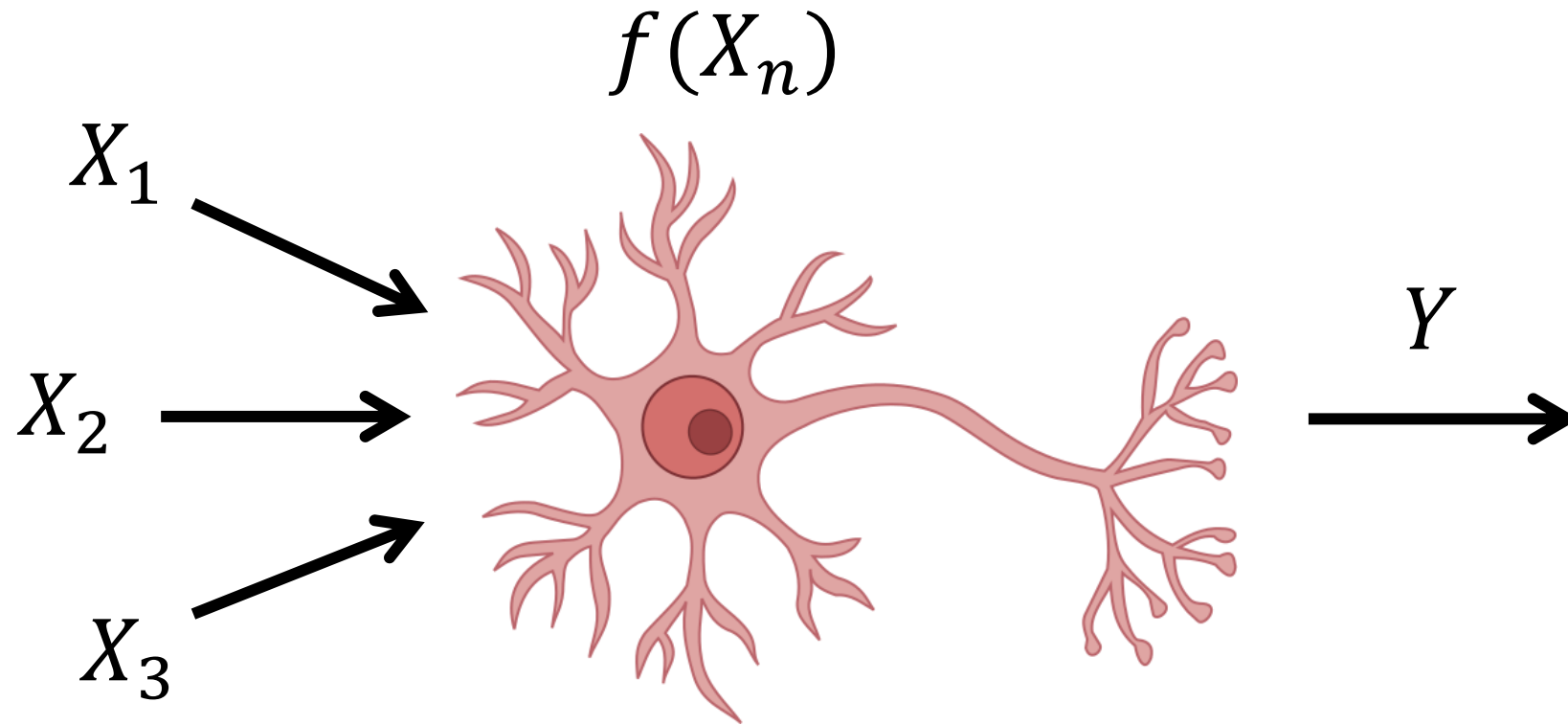
Studying network organisation and function in *in vitro* neuronal networks

*BioRTC/Simons Computational
Neuroscience Program 2026*

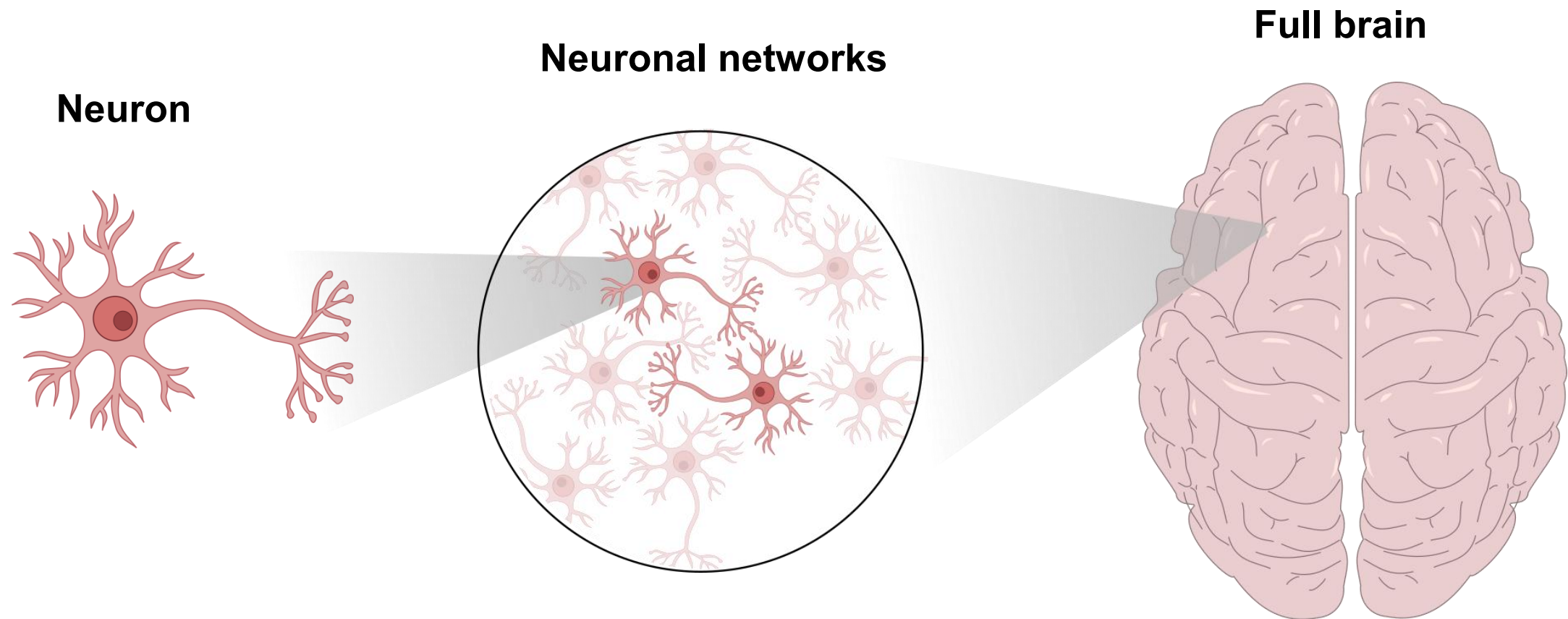
Laboratory of Biosensors and Bioelectronics, ETH Zürich
Giulia Amos



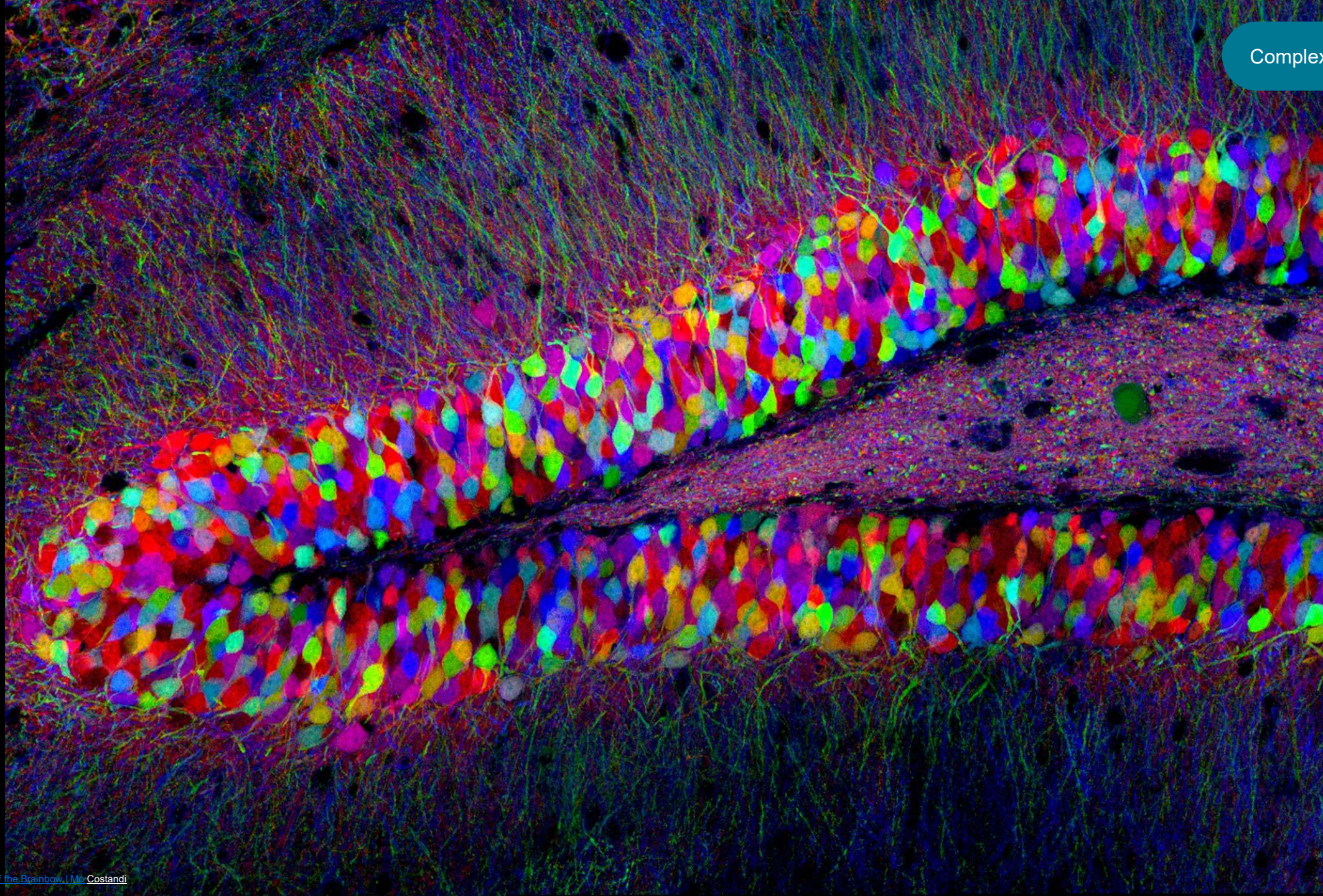
Neurons are the building blocks of the brain



Neurons are the building blocks of the brain





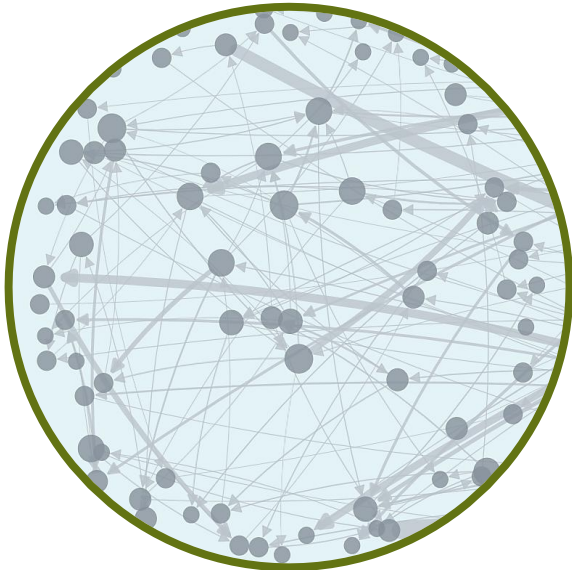


Experimental and statistical tools to study neuronal networks

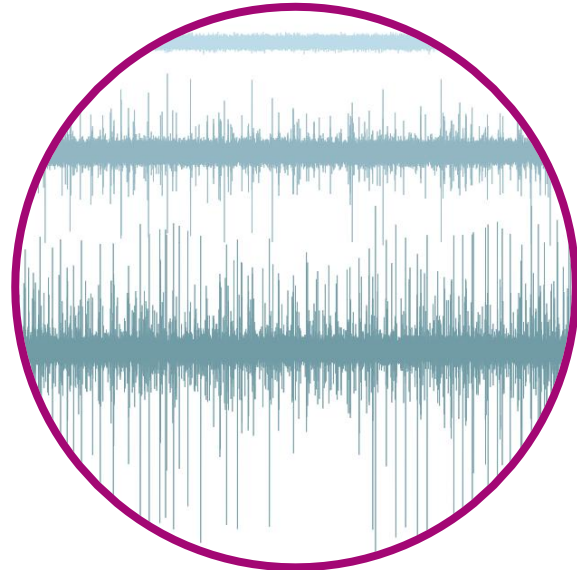
A complex system is highly composite, and interactions between sub-units give rise to rich collective behaviour. If brain function emerges from interactions between neurons, then the **organisation of those interactions** must matter!

We can study this using...

Statistical Tools



Network Analysis



Information Theory

Experimental Tools



***In vitro* Platform**



Electrophysiology

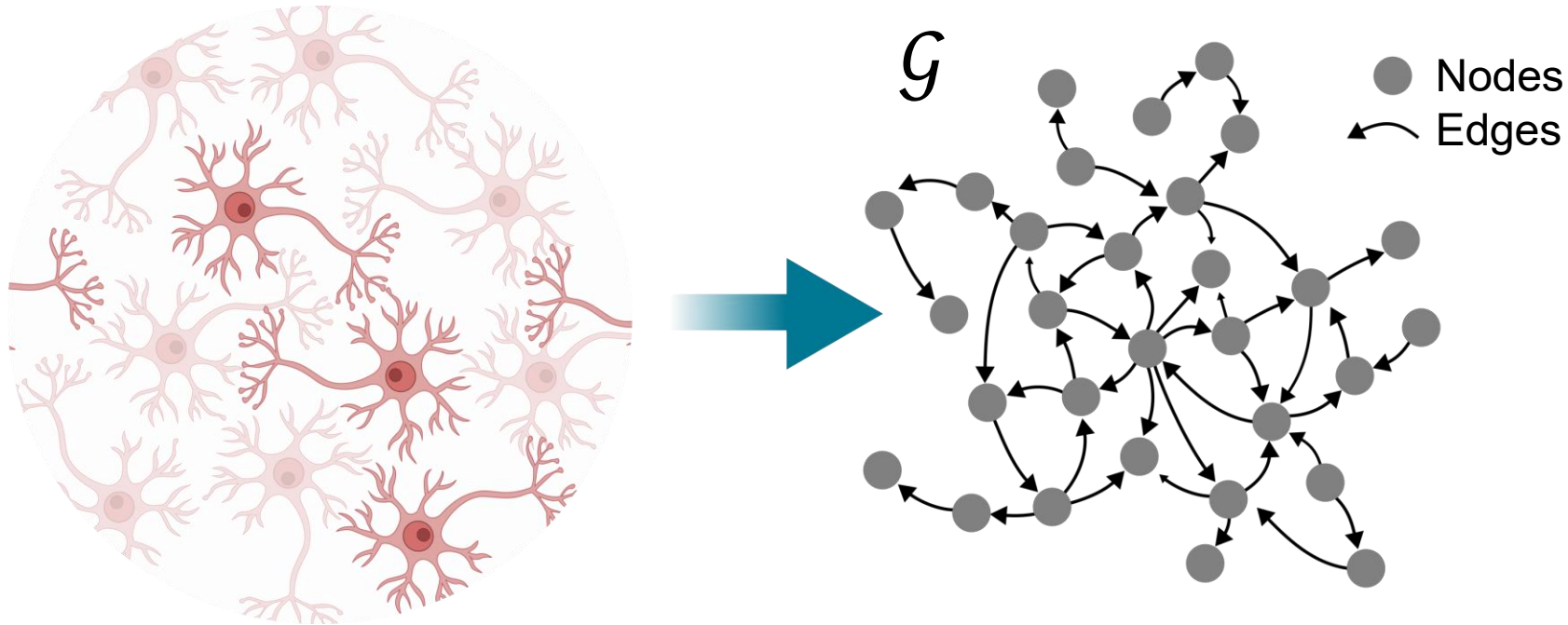
How can we describe and analyse interactions between neurons?

How can we describe and analyse interactions between neurons?

Networks are the skeletons of complex systems and can be studied using network theory.

Network neuroscience describes interactions between neurons

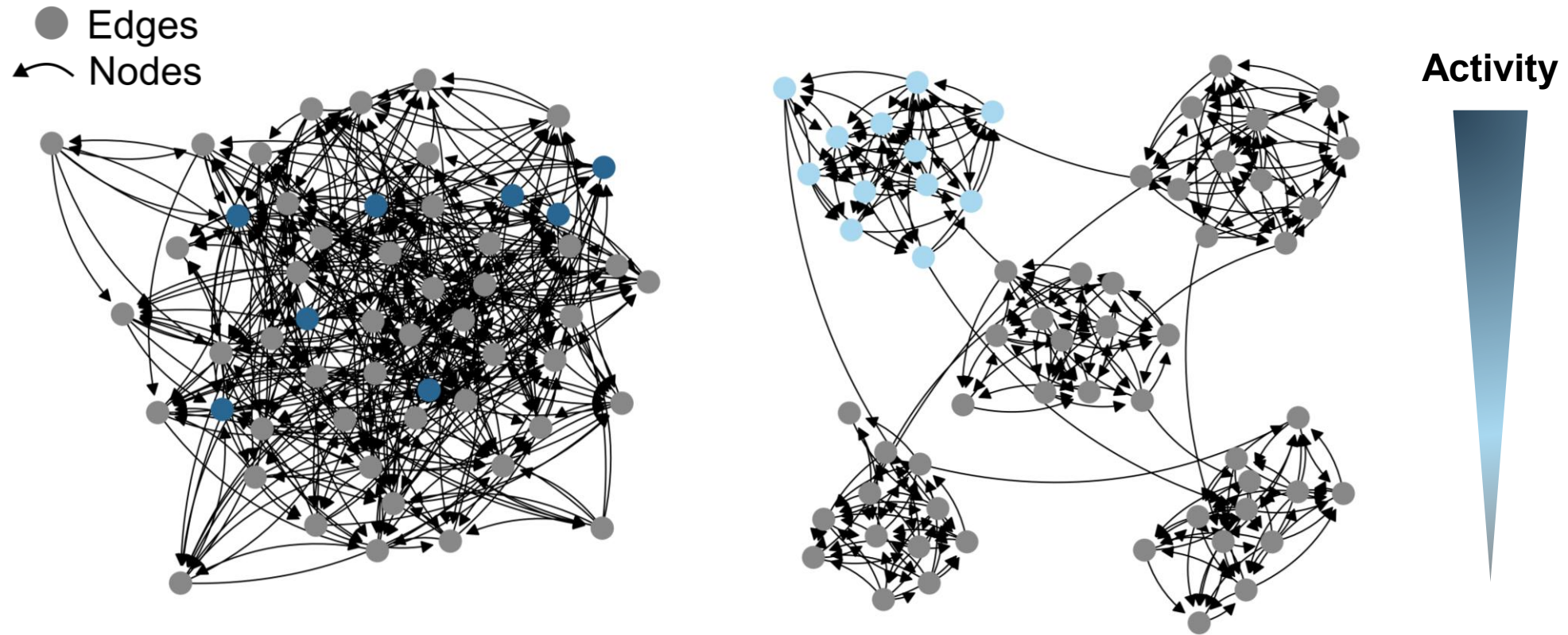
A network removes all unnecessary information and considers **only relevant features** that relate to the system's global dynamics to describe the **structure of information flow**.

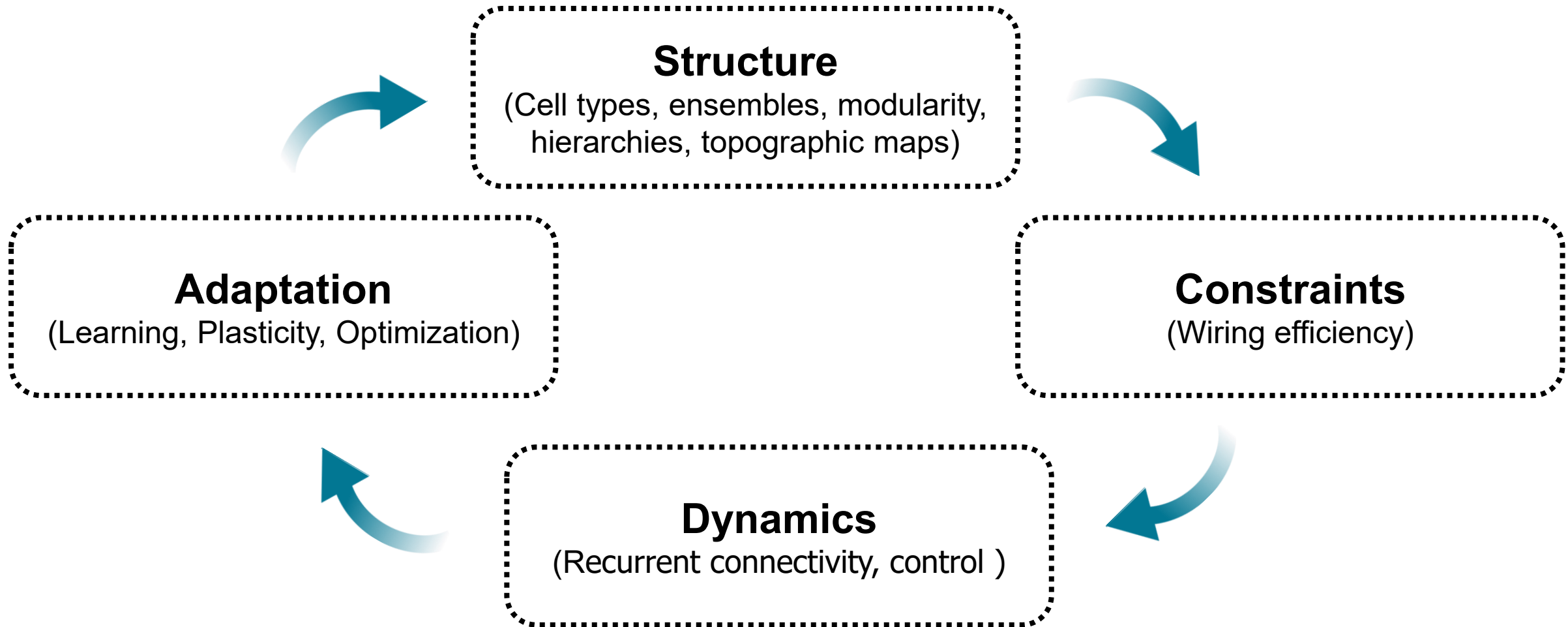


$$N = \{u, v, w \dots\}$$

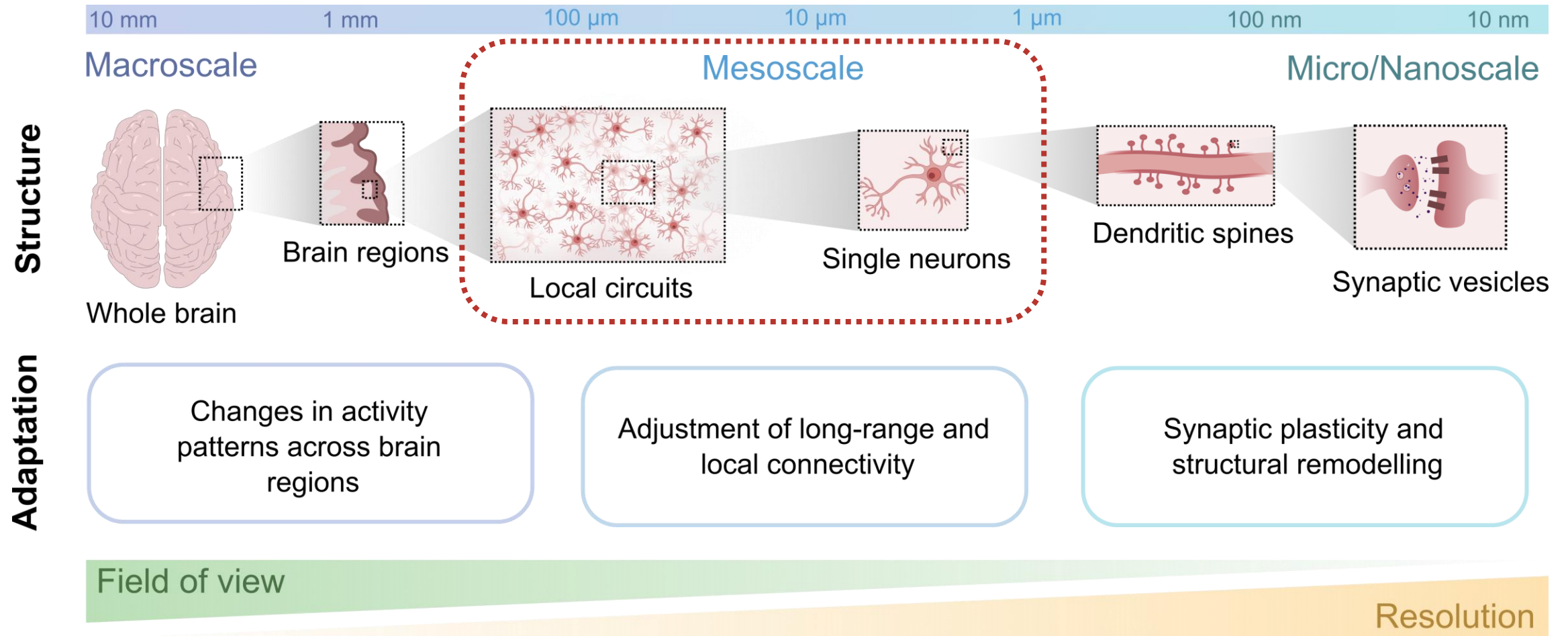
$$E = \{e = (u, v)\}, u, v \in \mathcal{V}$$

Network structure shapes how signals propagate and combine





The brain is organised at every scale



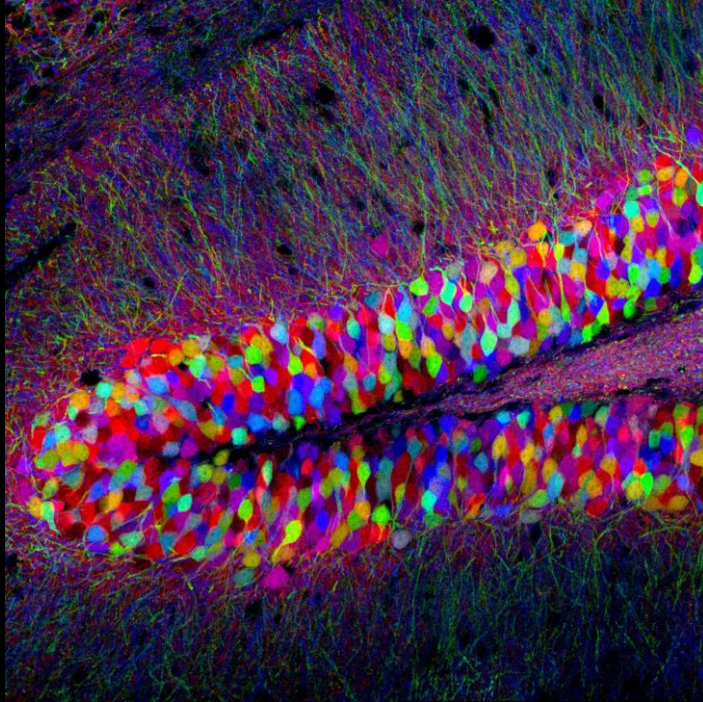
How can we study mesoscale network organisation in a controlled environment with high spatiotemporal resolution?

How can we study mesoscale network organisation in a controlled environment with high spatiotemporal resolution?

In vitro cultures offer a experimental platform to longitudinally study network reorganisation.

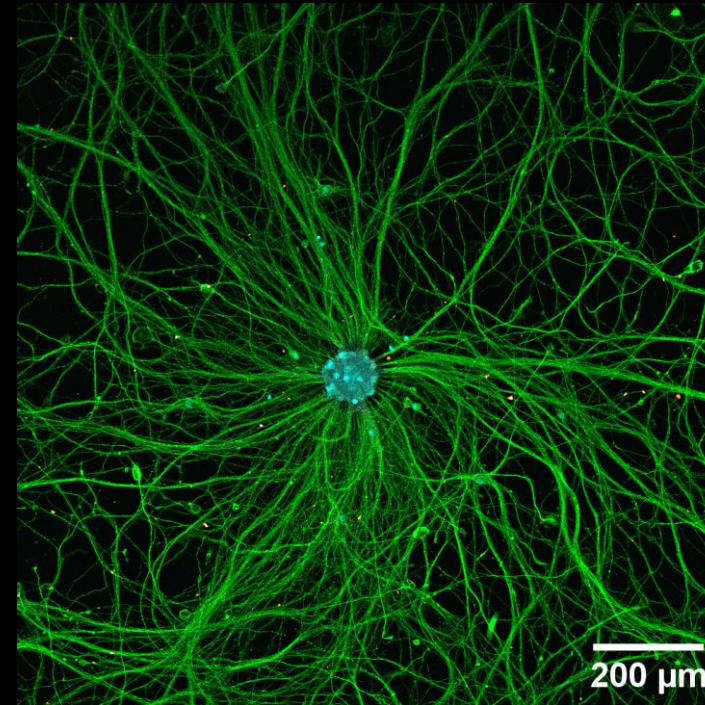
A bottom-up approach to neuroscience

in vivo



- Highly complex, even at smaller scales
- Physiologically relevant context

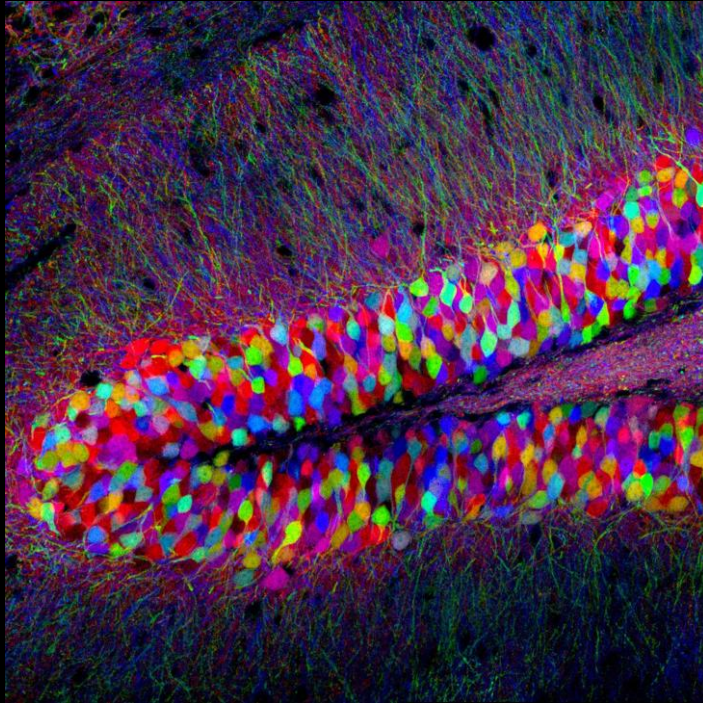
in vitro



- Controlled experimental environment
- Highly tunable system

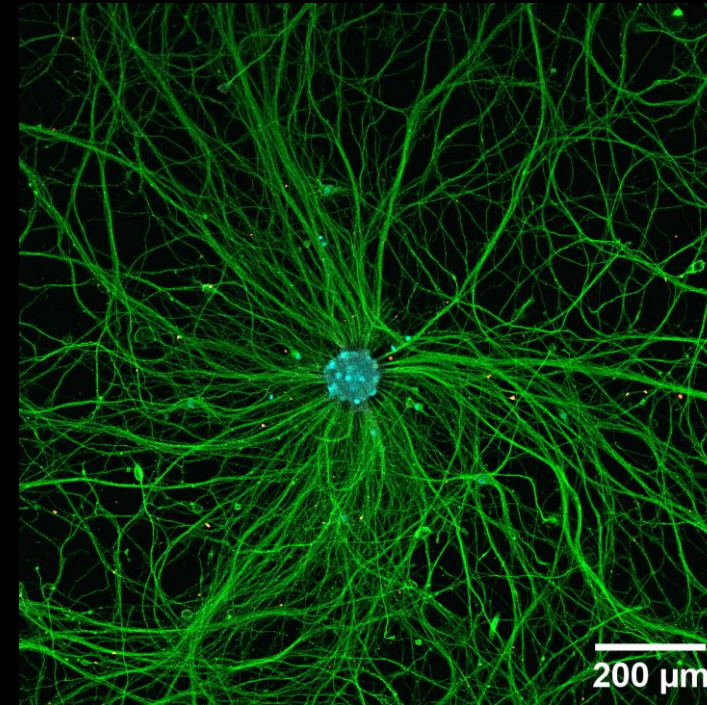
A bottom-up approach to neuroscience

in vivo



- Function-guided maturation
- Native connectivity

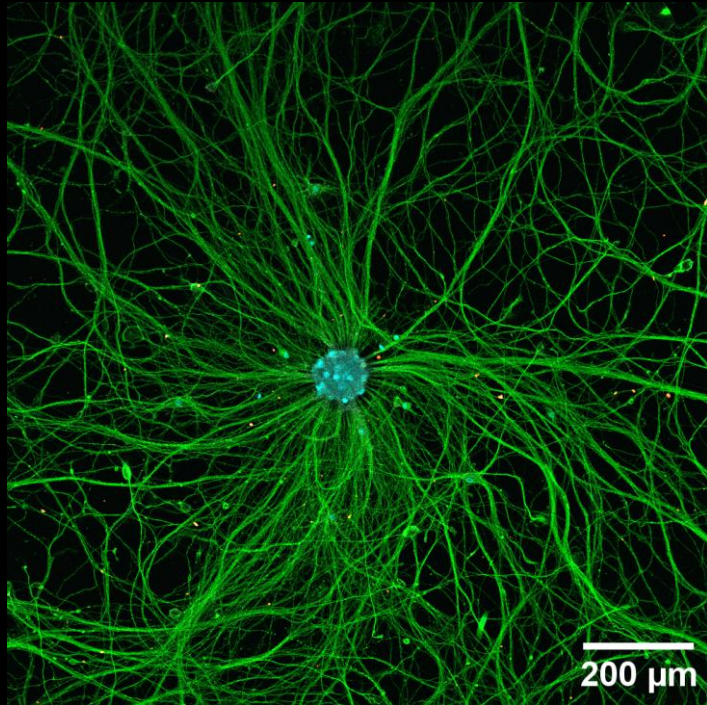
in vitro



- Reduced exposure to developmental cues and growth factors
- «Regrown» system, often without input

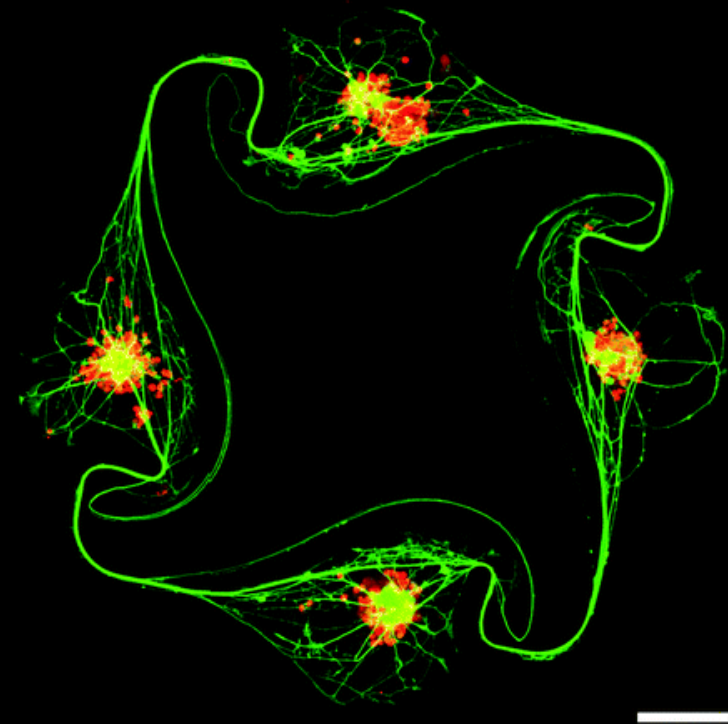
Microstructures can topologically constrain network topography

Random growth



Traditional cultures have random connectivity and limited control

Topologically constrained



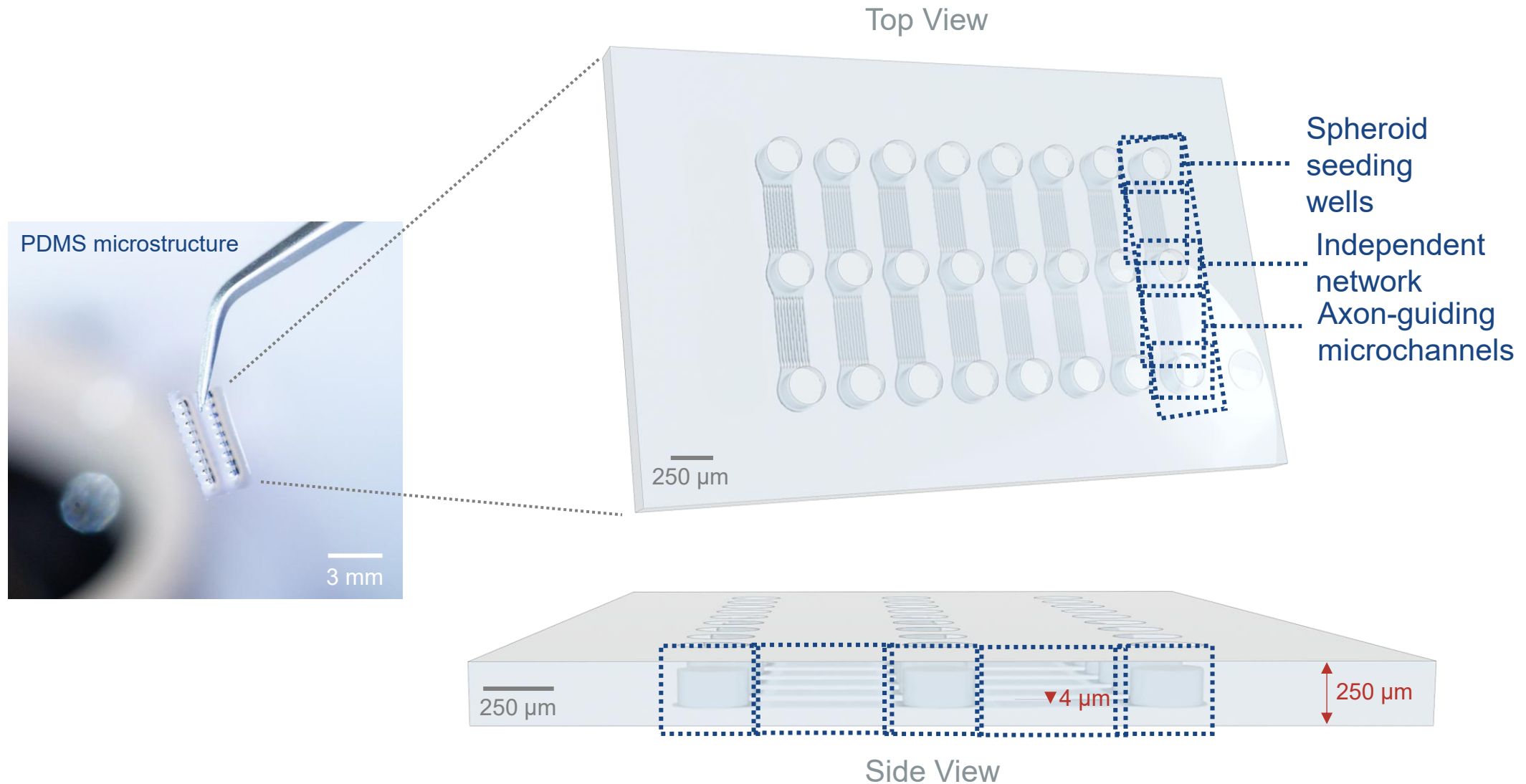
Microstructures can topologically constrain networks to mimic in vitro circuits

Microstructures can topologically constrain network topography

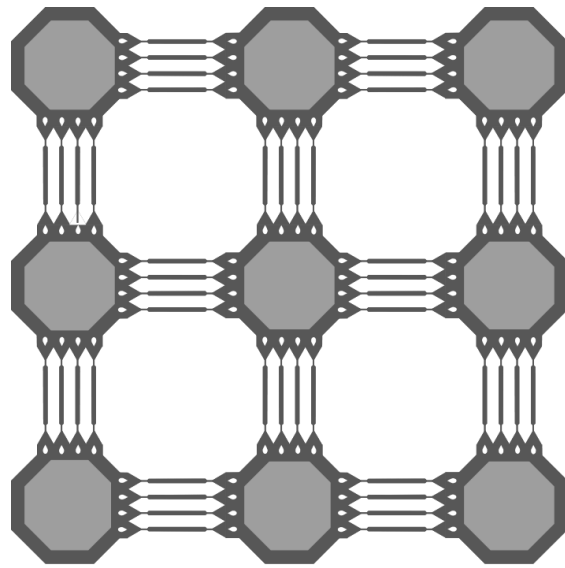
Polydimethylsiloxane (PDMS)
microstructure

3 mm

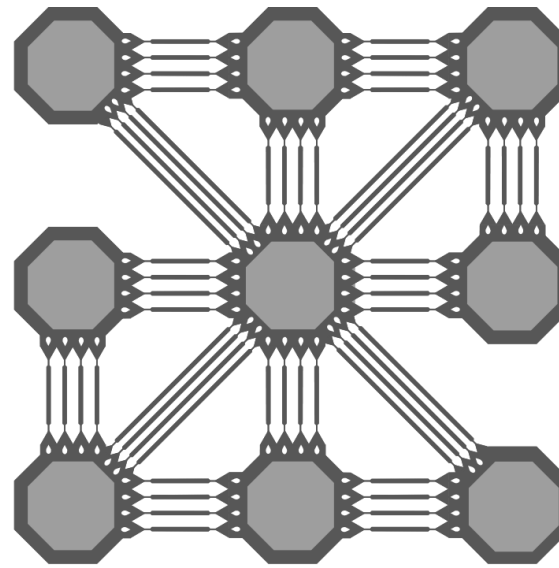
Microstructures can topologically constrain network topography



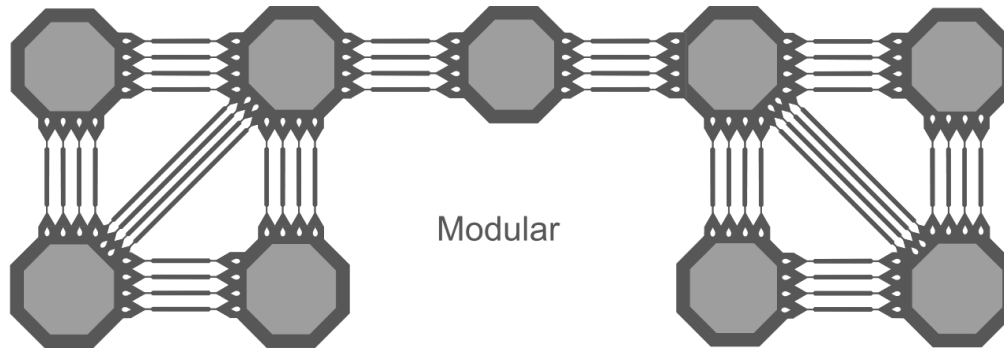
Example: Growth of spheroids in different microstructures



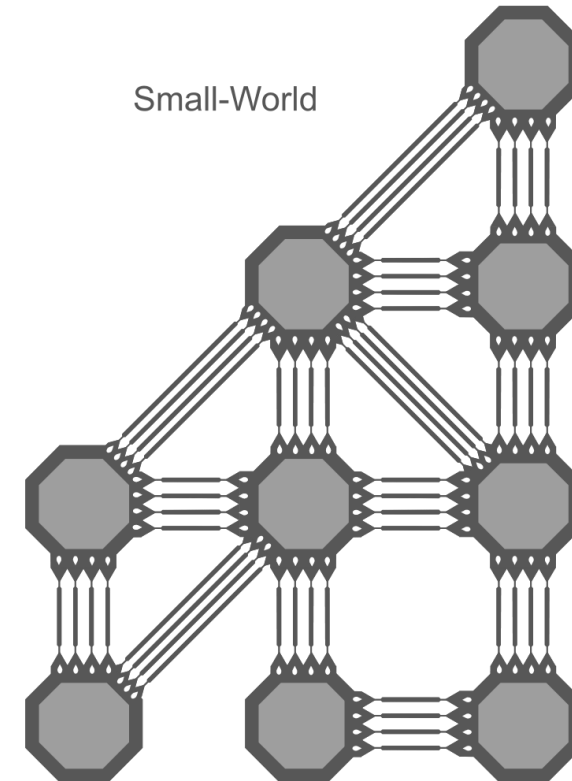
Lattice



Barabasi-Albert



Modular



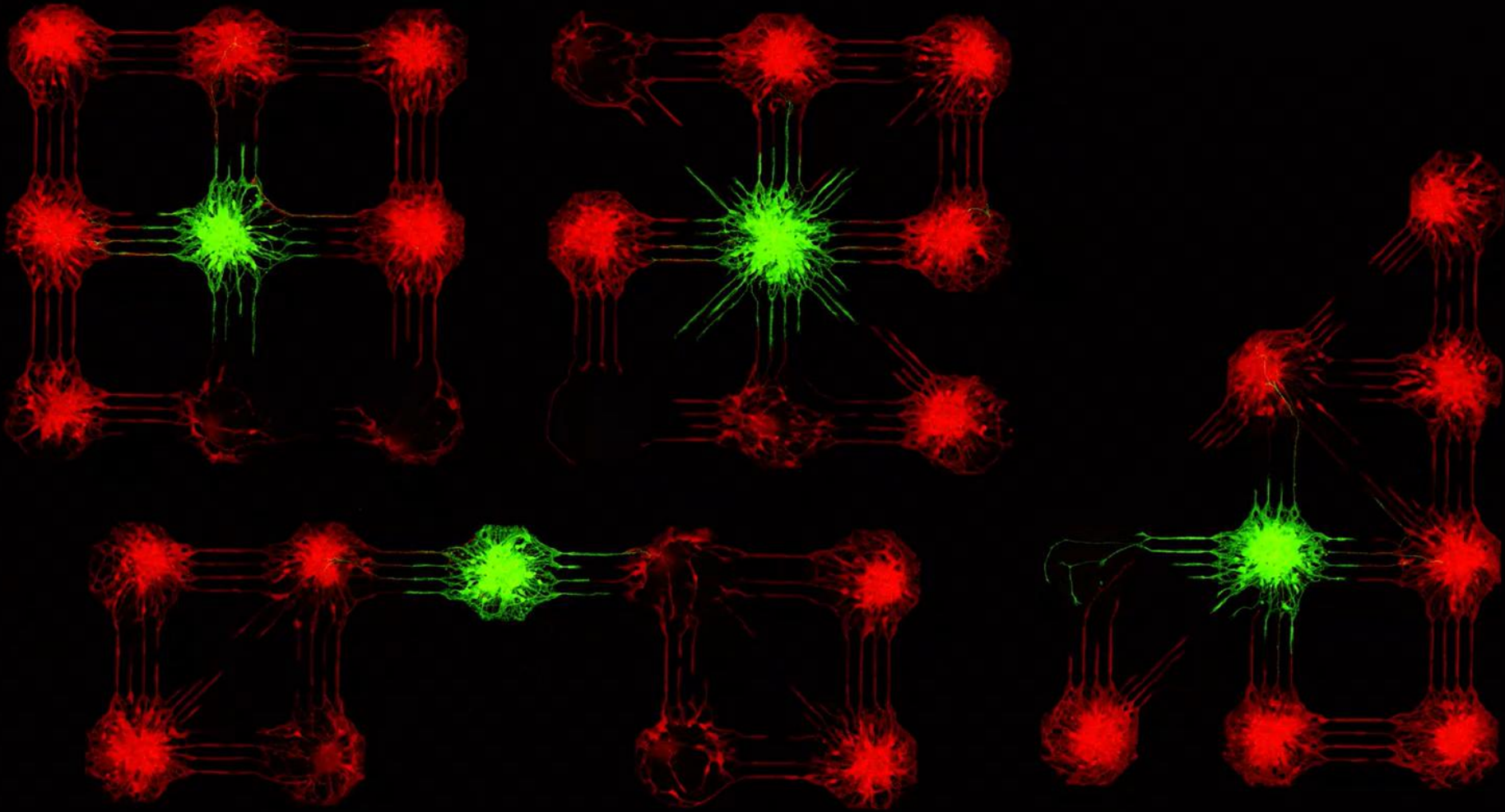
Small-World

DIV0-10

GFP Neurons

RFP Neurons

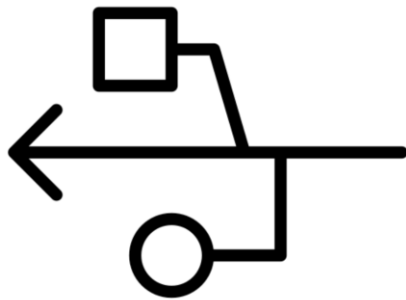
Engineered
Neuronal Networks



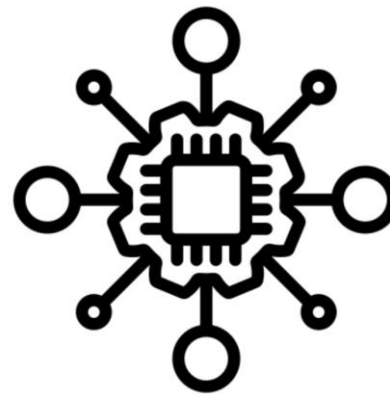
200 μm

Microstructures can topologically constrain network topography

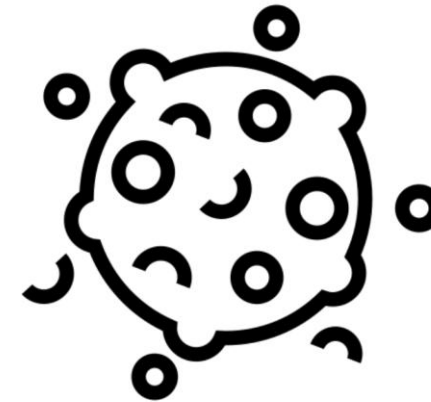
This enables to ask how network structure shapes...



Integration / segregation



Computation



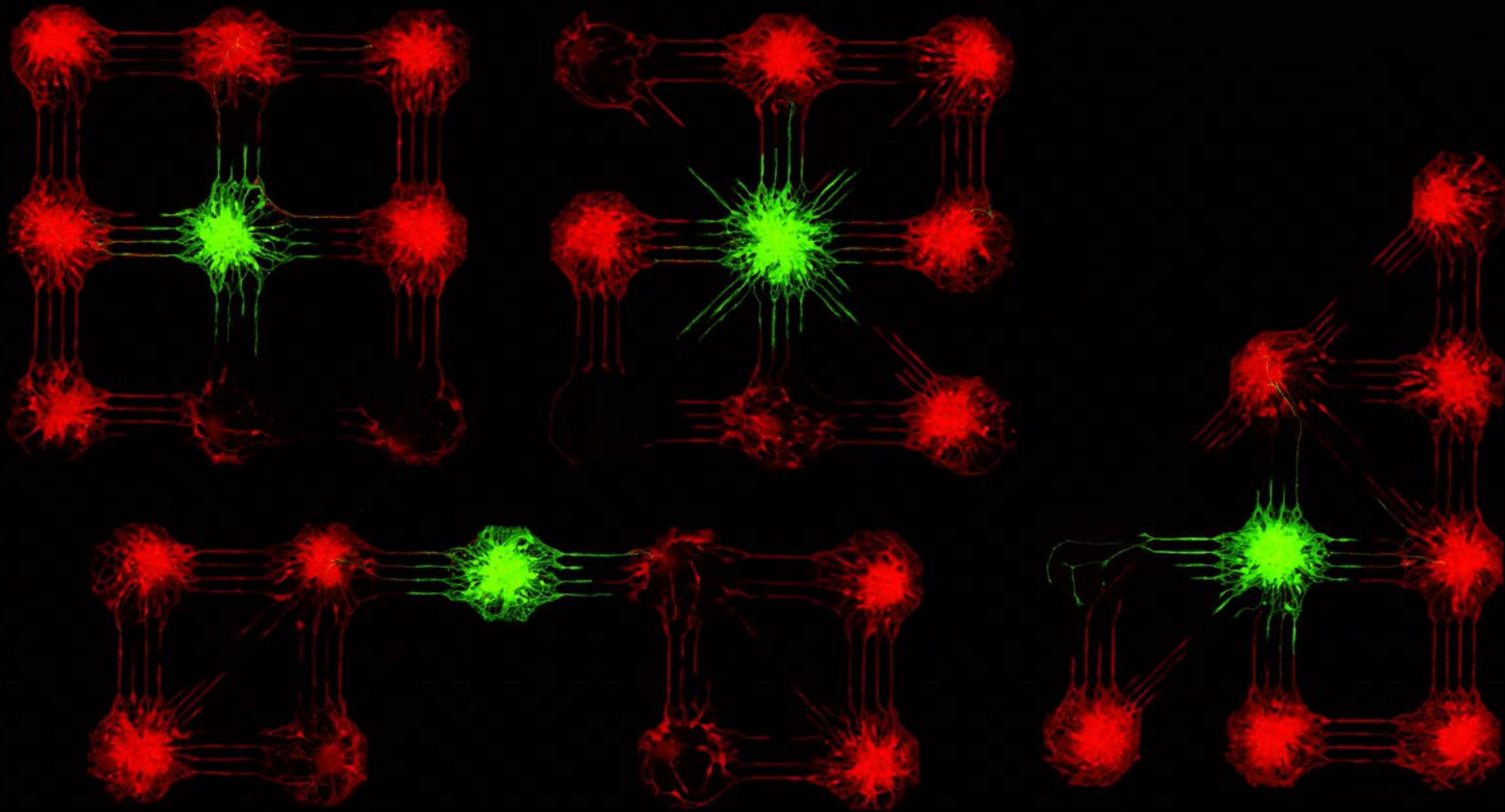
Disease progression

DIV0-10

GFP Neurons

RFP Neurons

Engineered
Neuronal Networks



200 μm

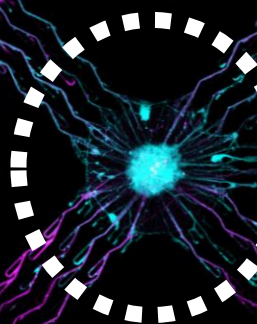
DIV12
Calcein

Engineered
Neuronal Networks

14 DIV

Engineered
Neuronal Networks

Disease
Induction



Human NGN2-Induced Neurons (GFP+)

Human NGN2-Induced Neurons (RFP+)



200 μ m

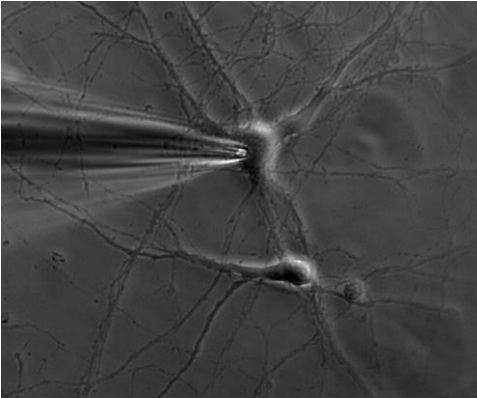
How can we record neuronal activity in these engineered neuronal networks?

How can we record neuronal activity in these engineered neuronal networks?

High-density microelectrode arrays offer high spatiotemporal resolution.

We collect data from neuronal populations at single-cell resolution

Functional readouts

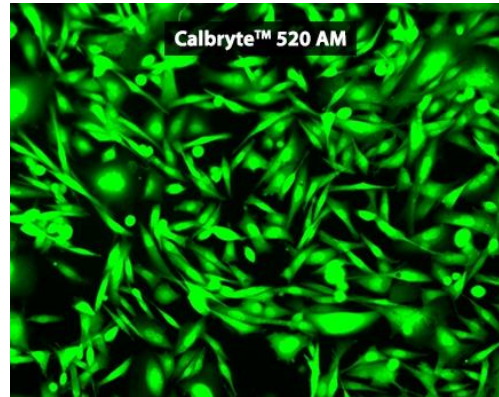


Patch Clamp

Single neuron

High temporal
resolution

High spatial
resolution

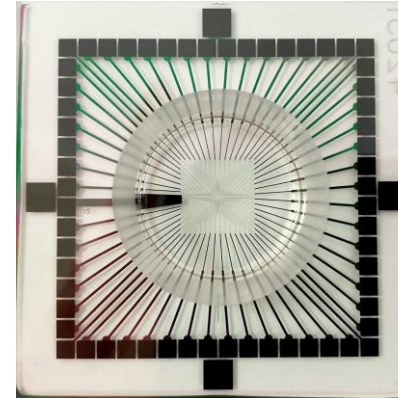


Calcium Imaging

Single neuron
and networks

Low temporal
resolution

High spatial
resolution



Microelectrode Arrays (MEAs)

Networks

High temporal
resolution

Low spatial
resolution



High-density MEAs

Networks

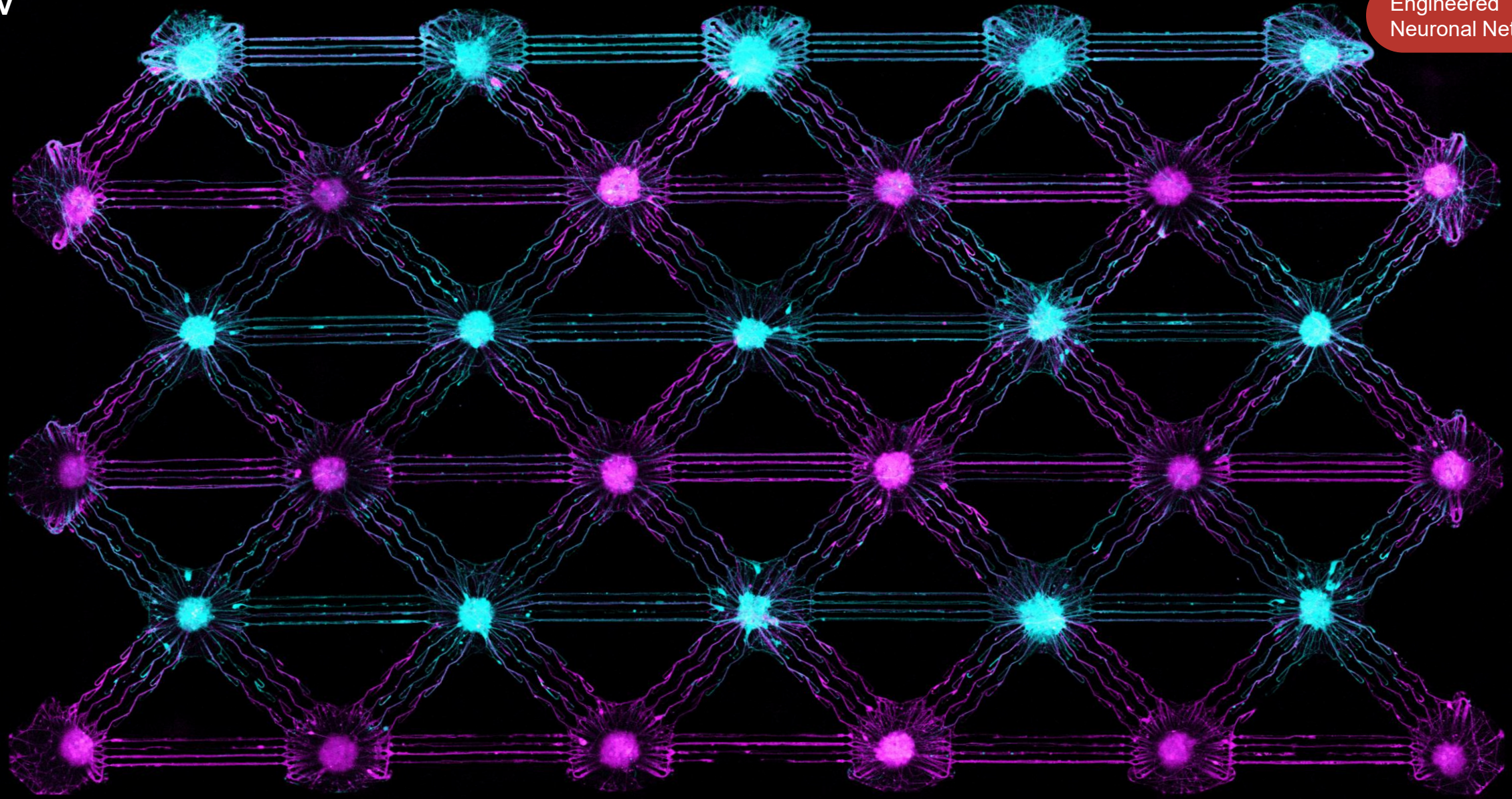
High temporal
resolution

High spatial
resolution

Neuronal activity *in vitro* can be recorded at multiple scales

Functional readouts



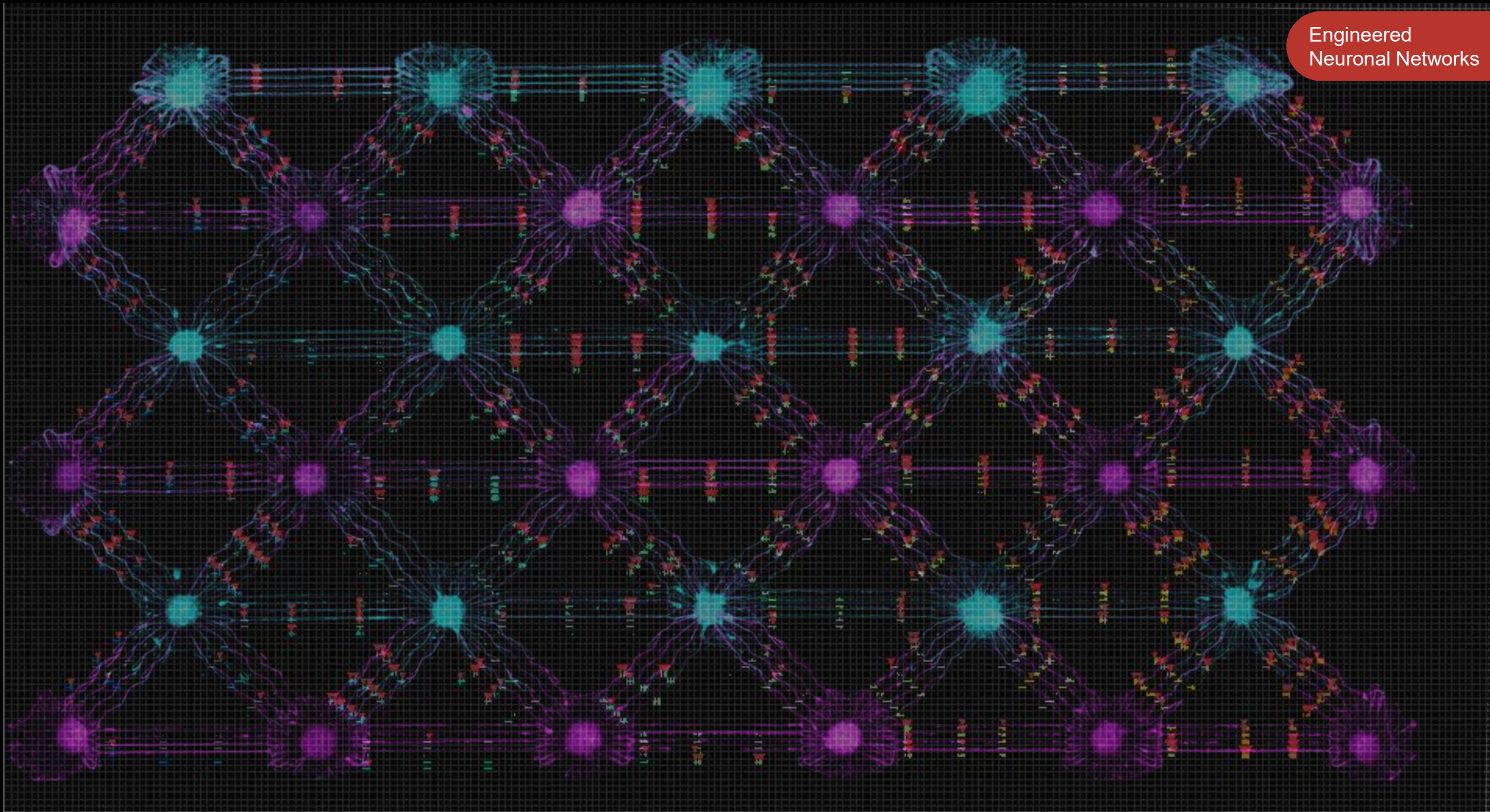


Human NGN2-Induced Neurons (GFP+)

Human NGN2-Induced Neurons (RFP+)



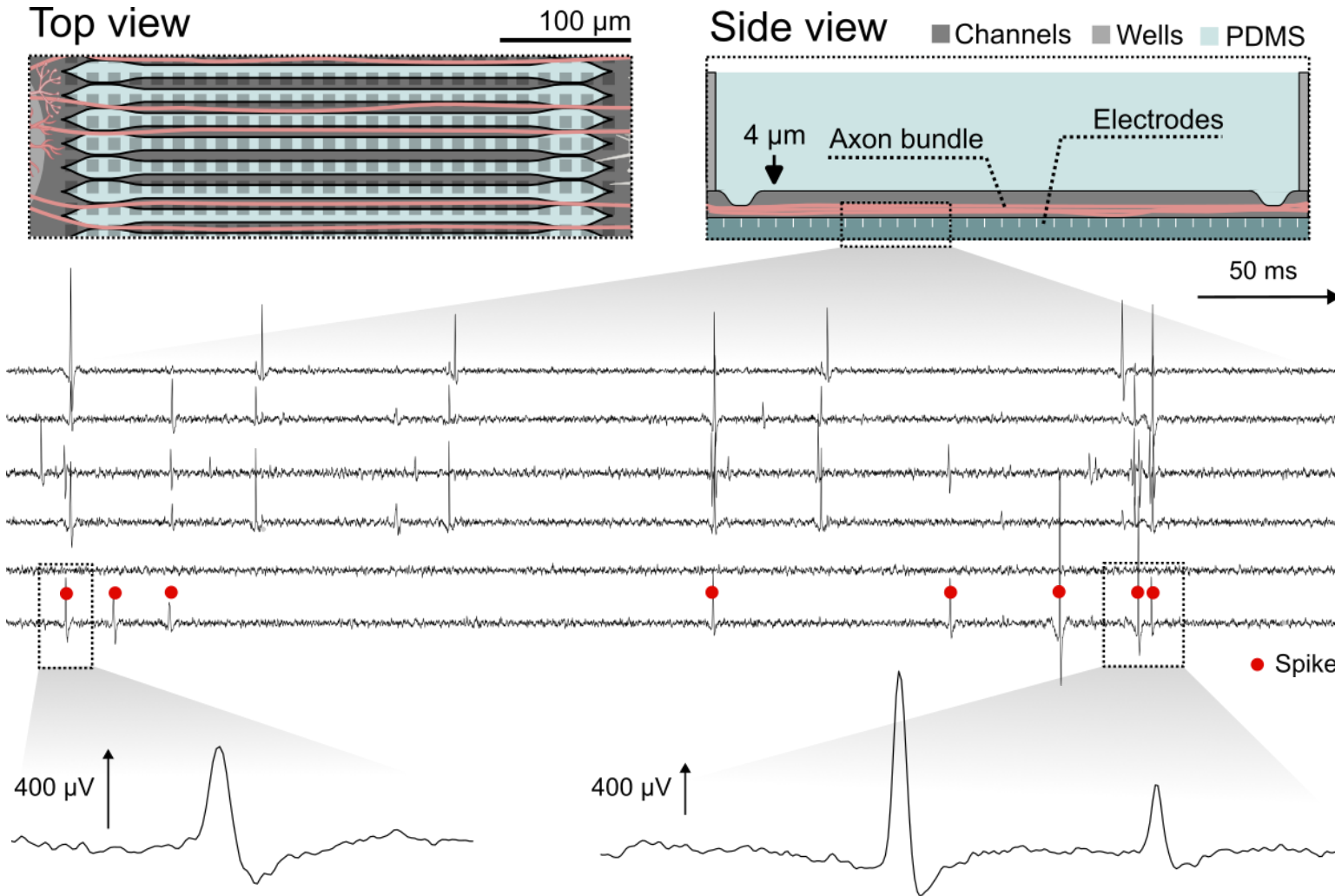
200 μ m



Human NGN2-Induced Neurons (GFP+)

Human NGN2-Induced Neurons (RFP+)

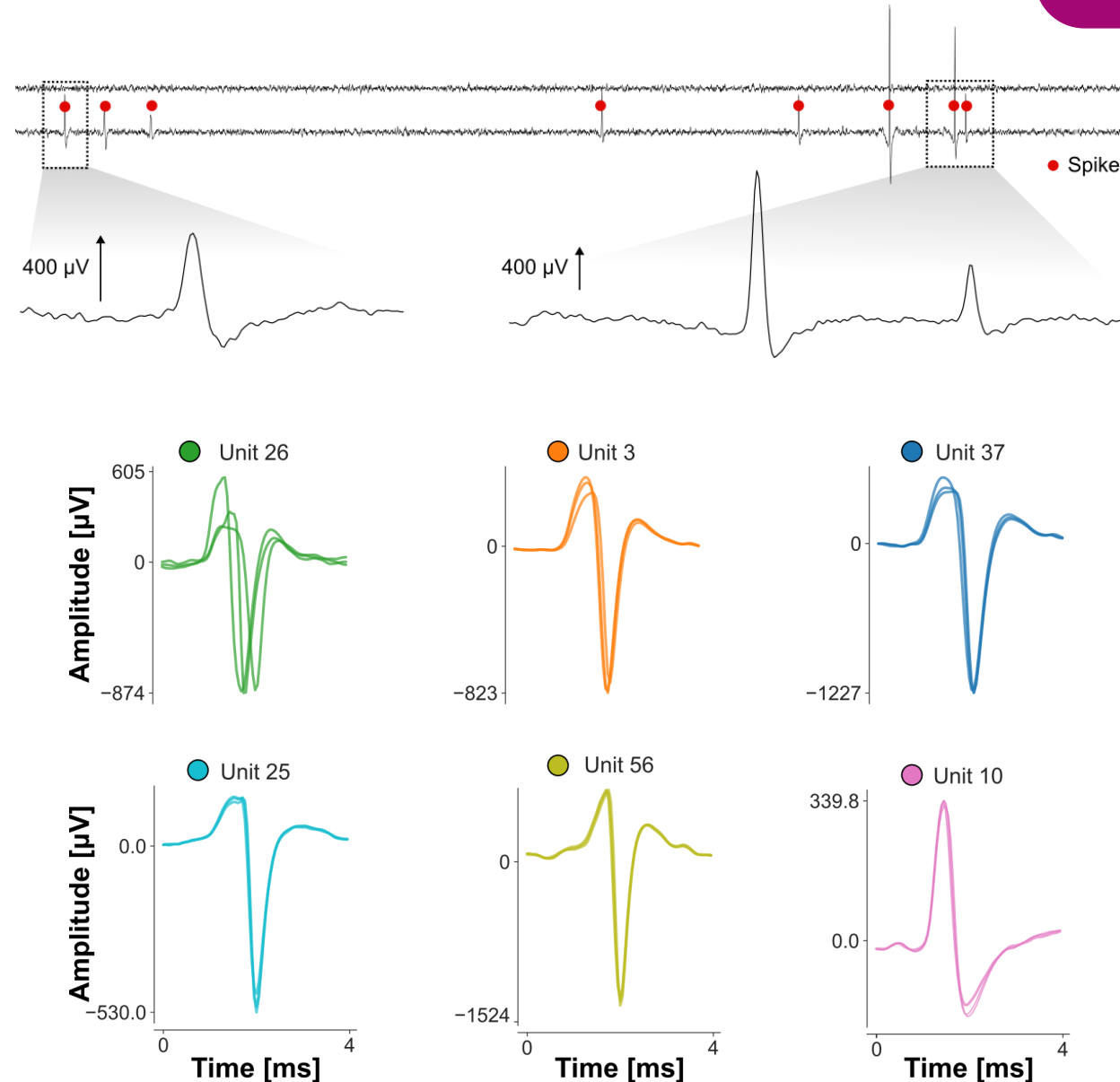
From voltage traces to single spike trains

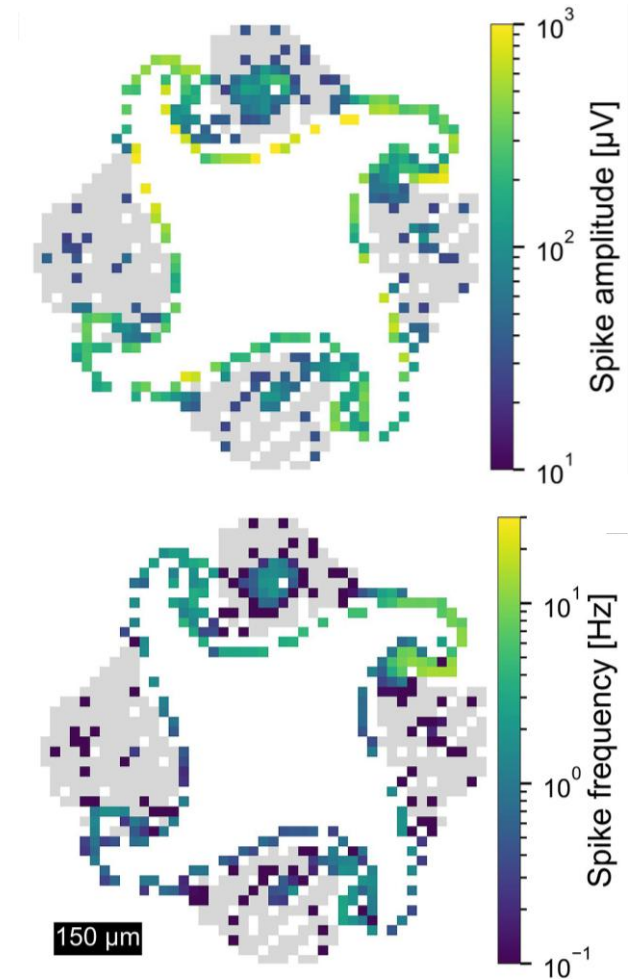
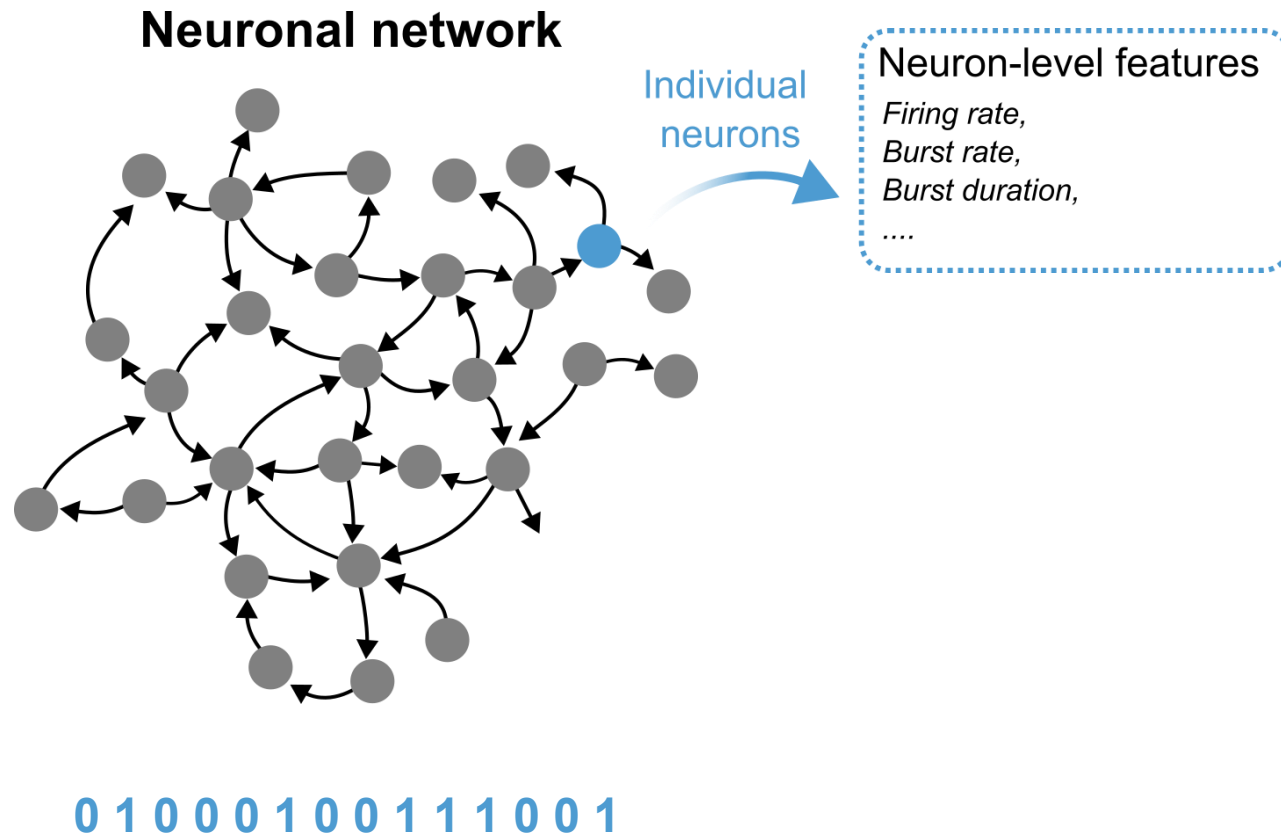


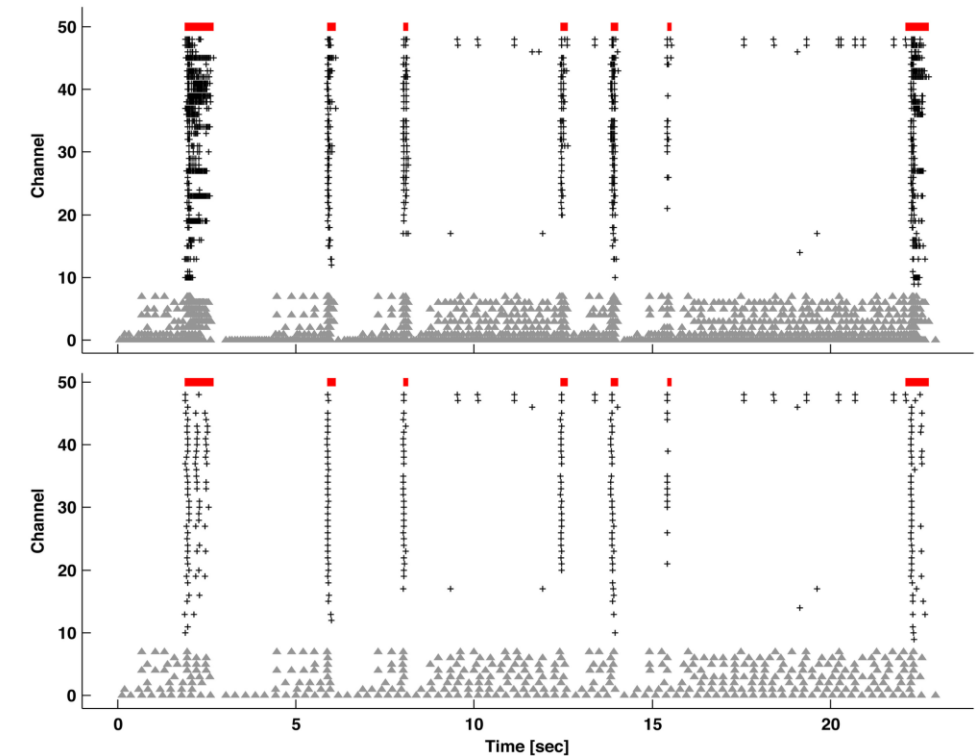
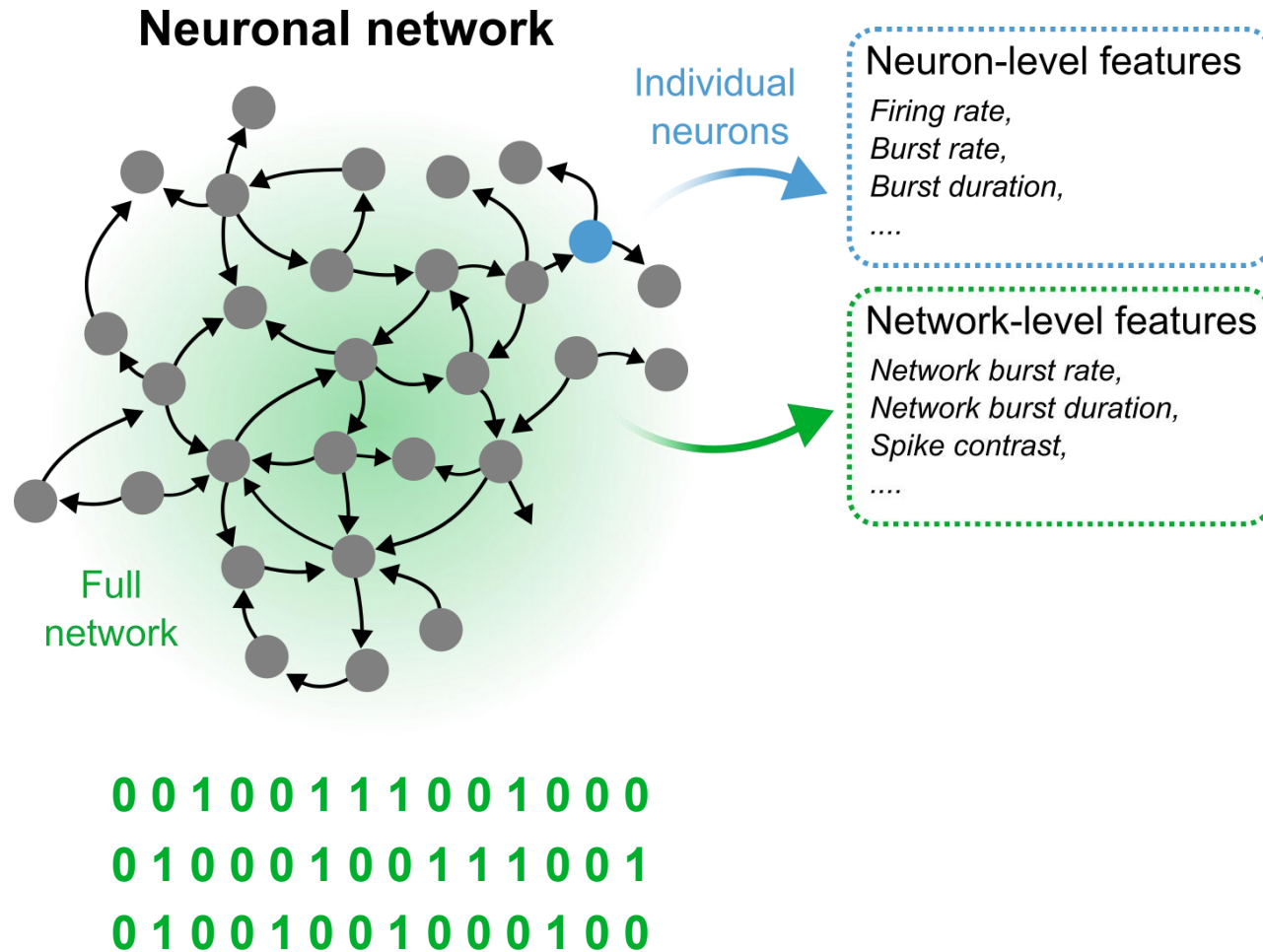
From voltage traces to single spike trains

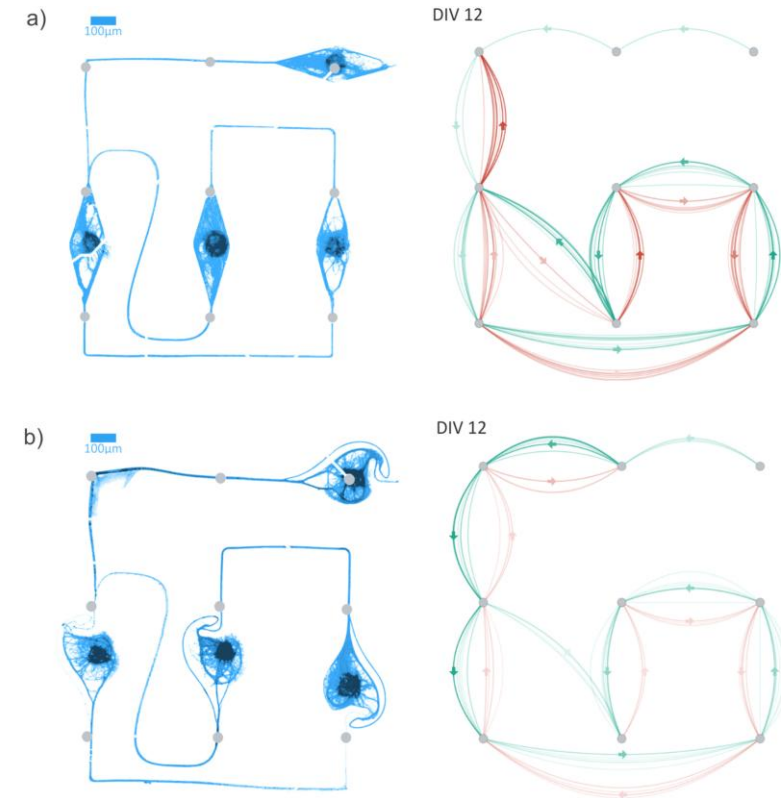
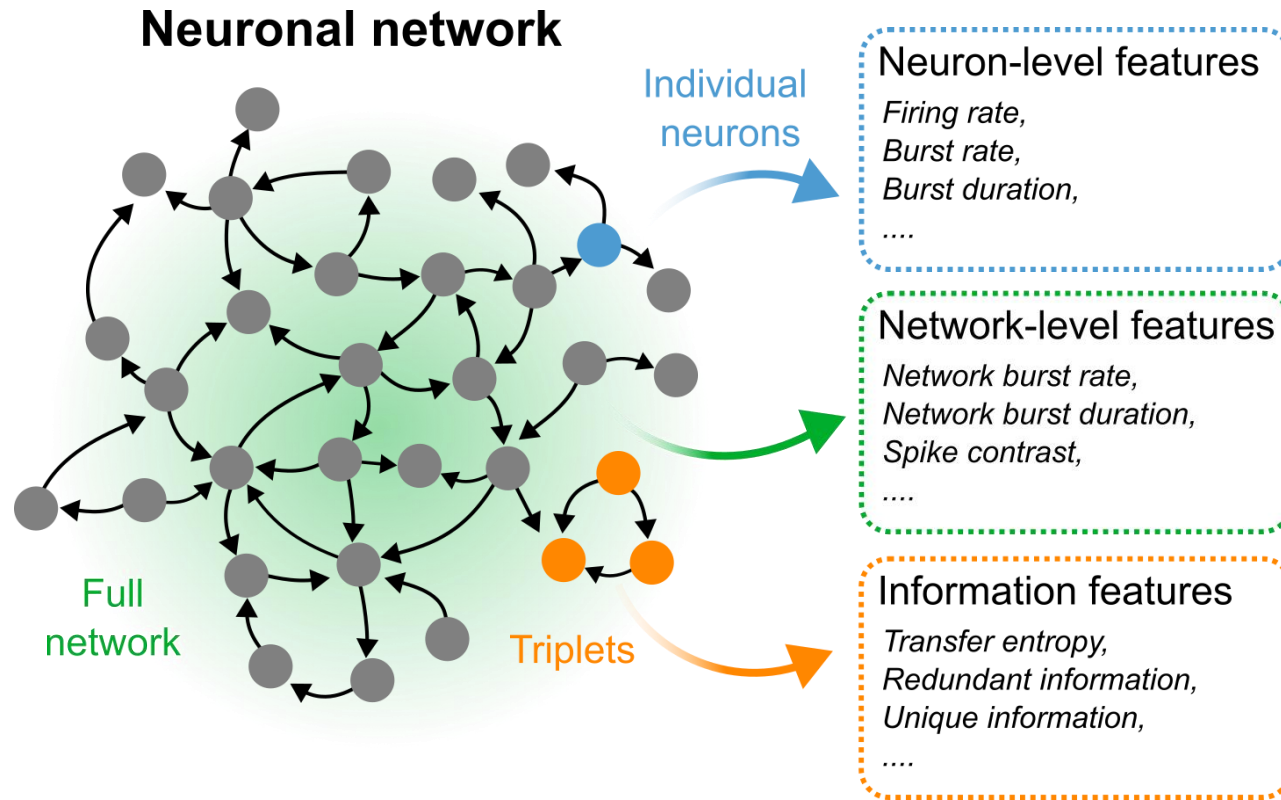
- Each electrode records from multiple nearby neurons, producing a **summed extracellular signal**
- Electrode placement affects **spike shape** and **amplitude**
- **Spike sorting** extracts individual neurons from recordings

→ This is the data you are going to get from me: **Spike times x neuron (binarized)**

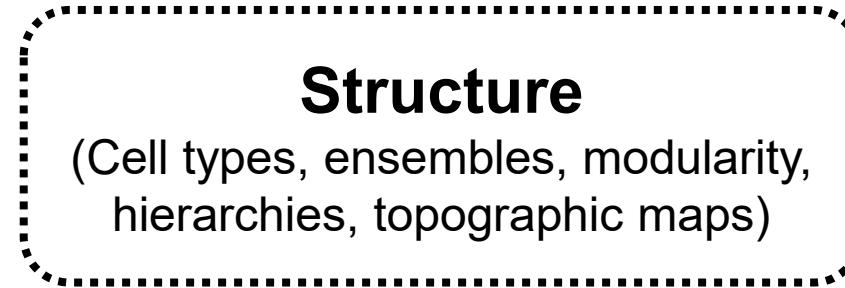








Pharmacology / disease models
Adaptive rewiring



Microstructures
Chemical gradients



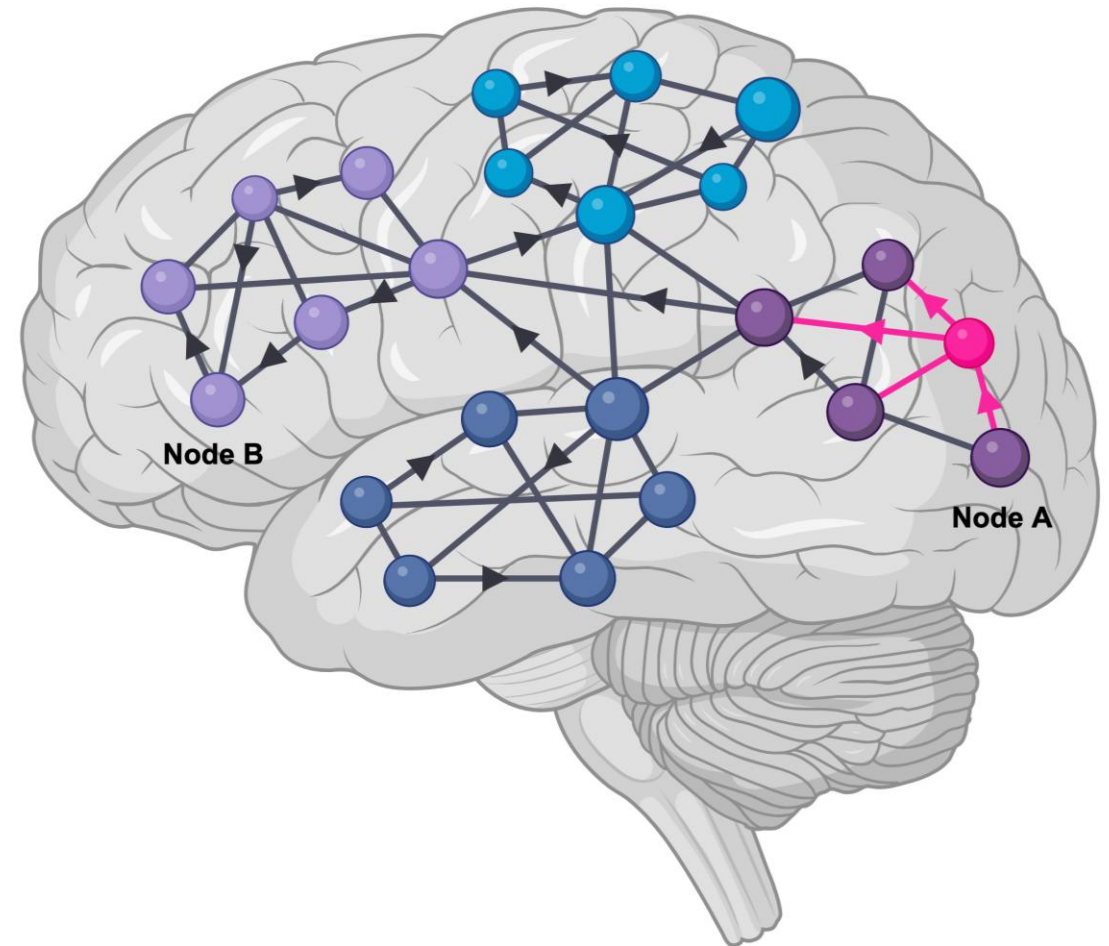
**Electrical / optogenetic
stimulation**
Plasticity protocols

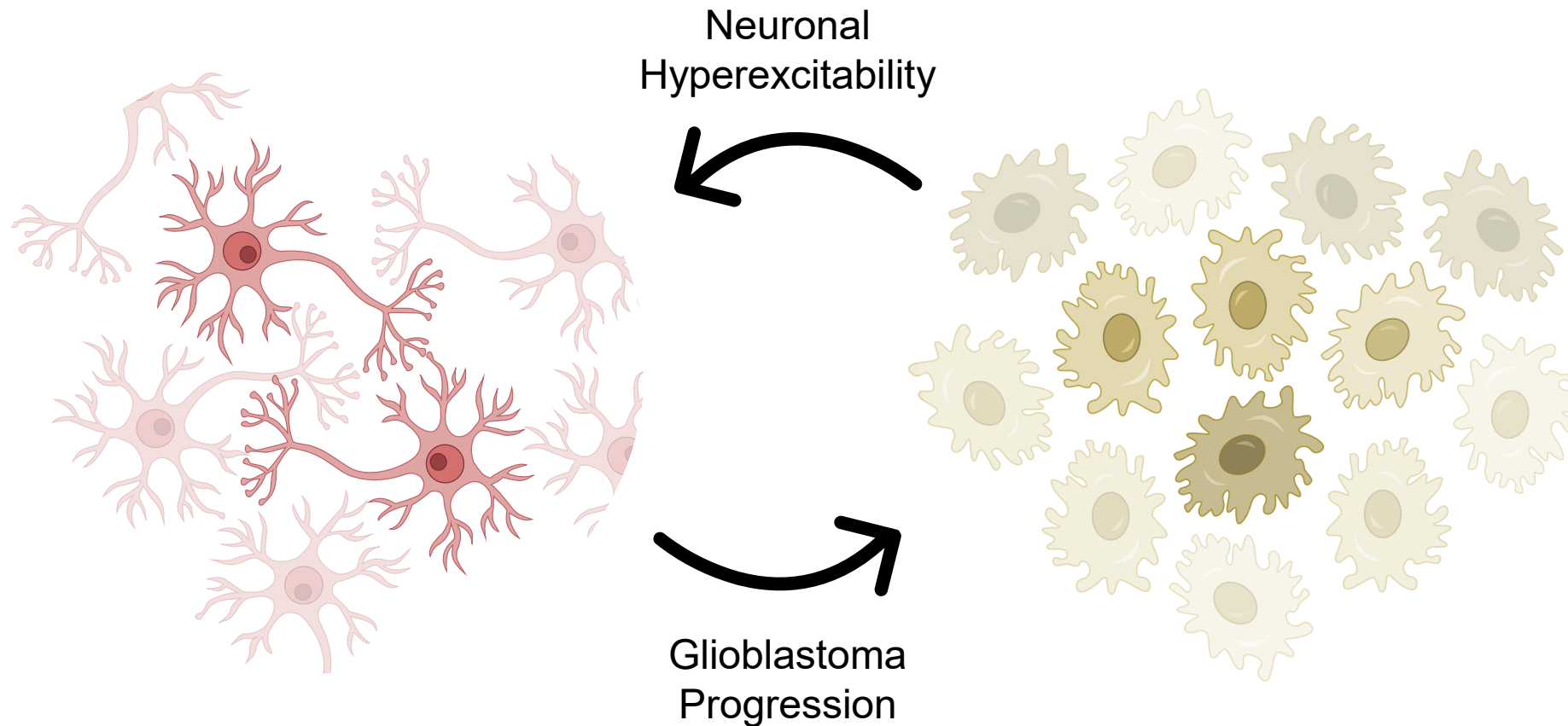


HD-MEA recordings
Network activity mapping

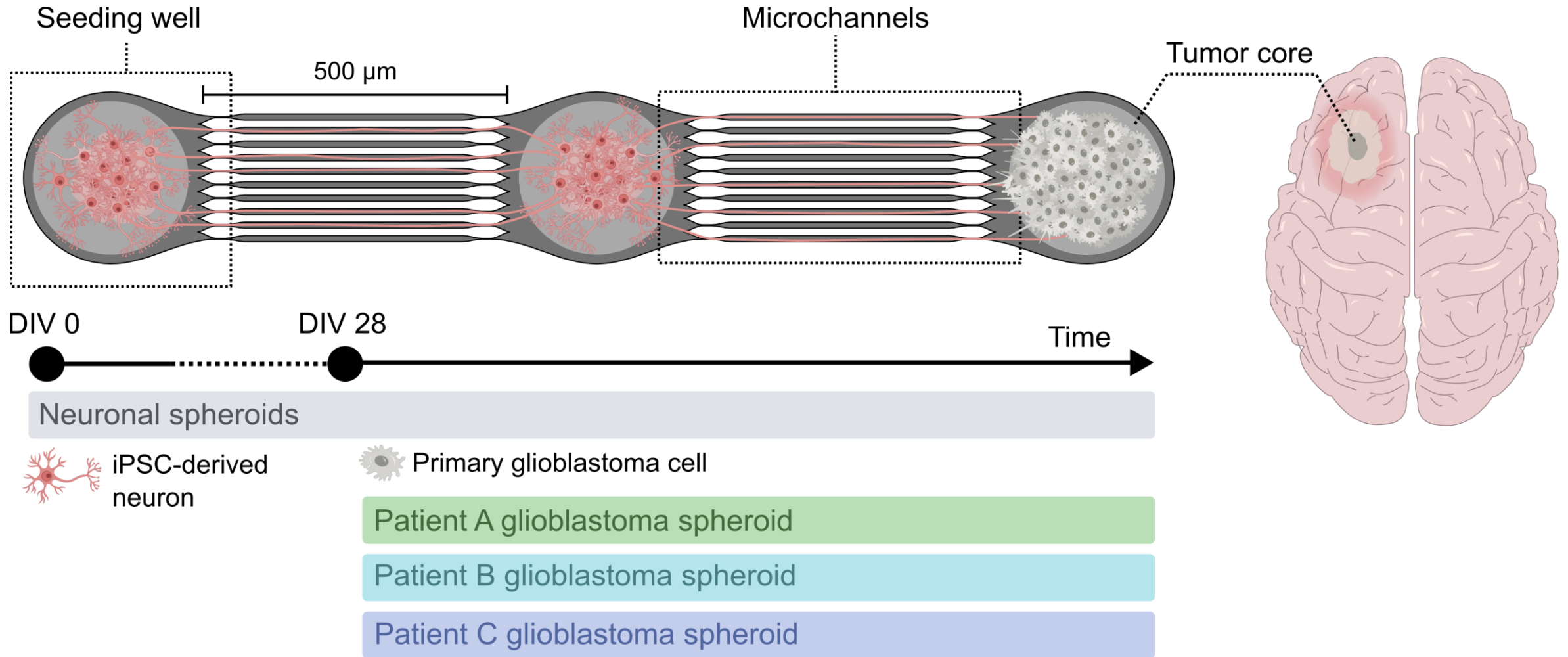
What is this useful for? 😊

- **Neurological diseases often begin in specific brain regions** and spread through networks
(e.g., toxic protein aggregates or altered neural activity)
- As pathology develops, the brain can **adapt and reroute information** through alternative pathways to maintain function
- With continued disease progression, pathology may reach **central hub regions of the network**, leading to persistent functional deficits

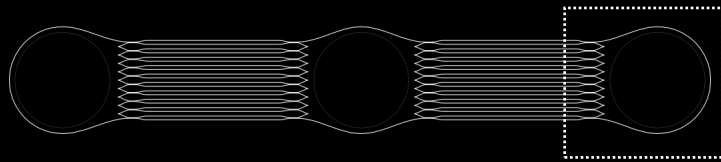




Glioblastoma cells are introduced at targeted locations in mature neuronal network

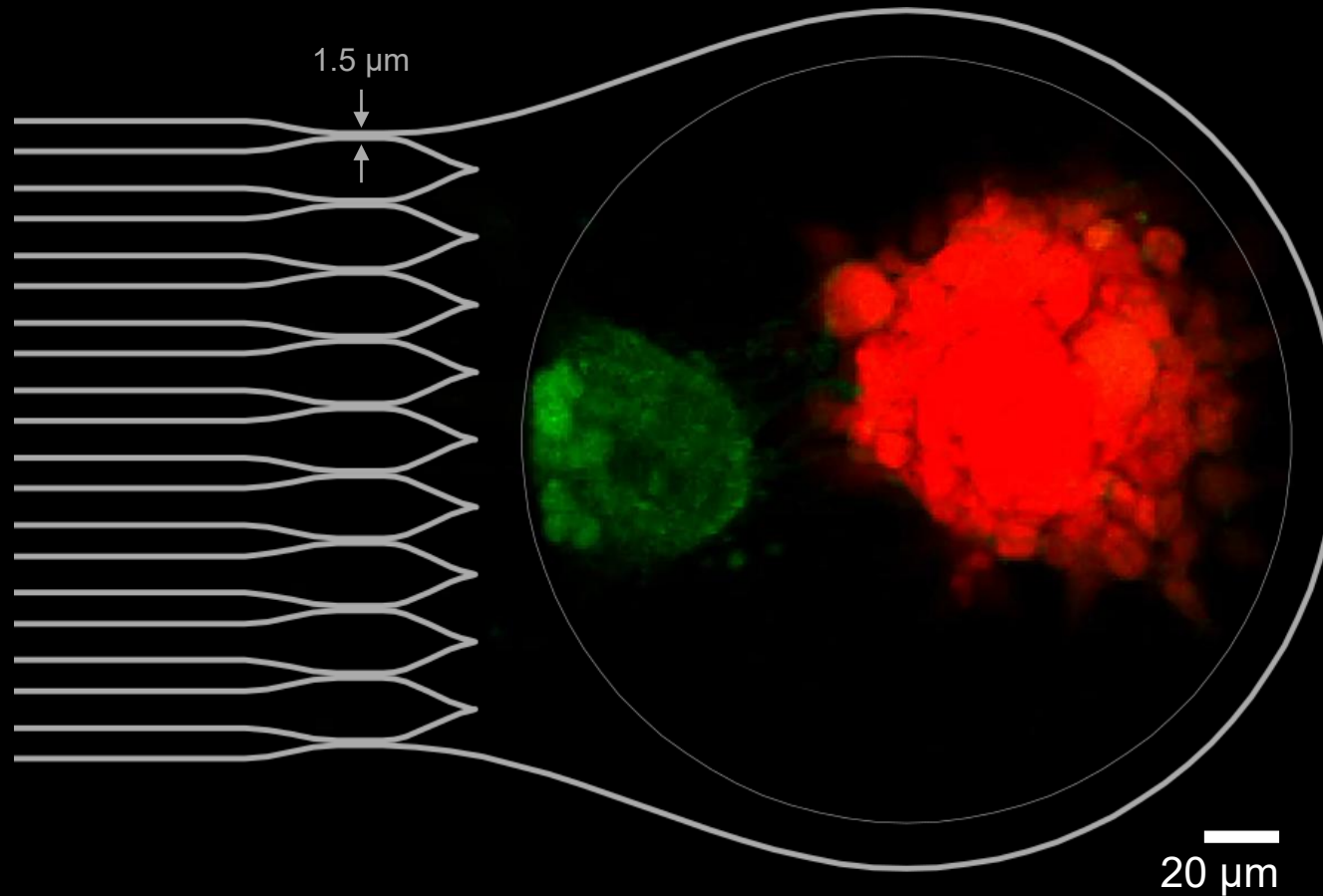


Glioblastoma cells approach and integrate into neuronal bundles

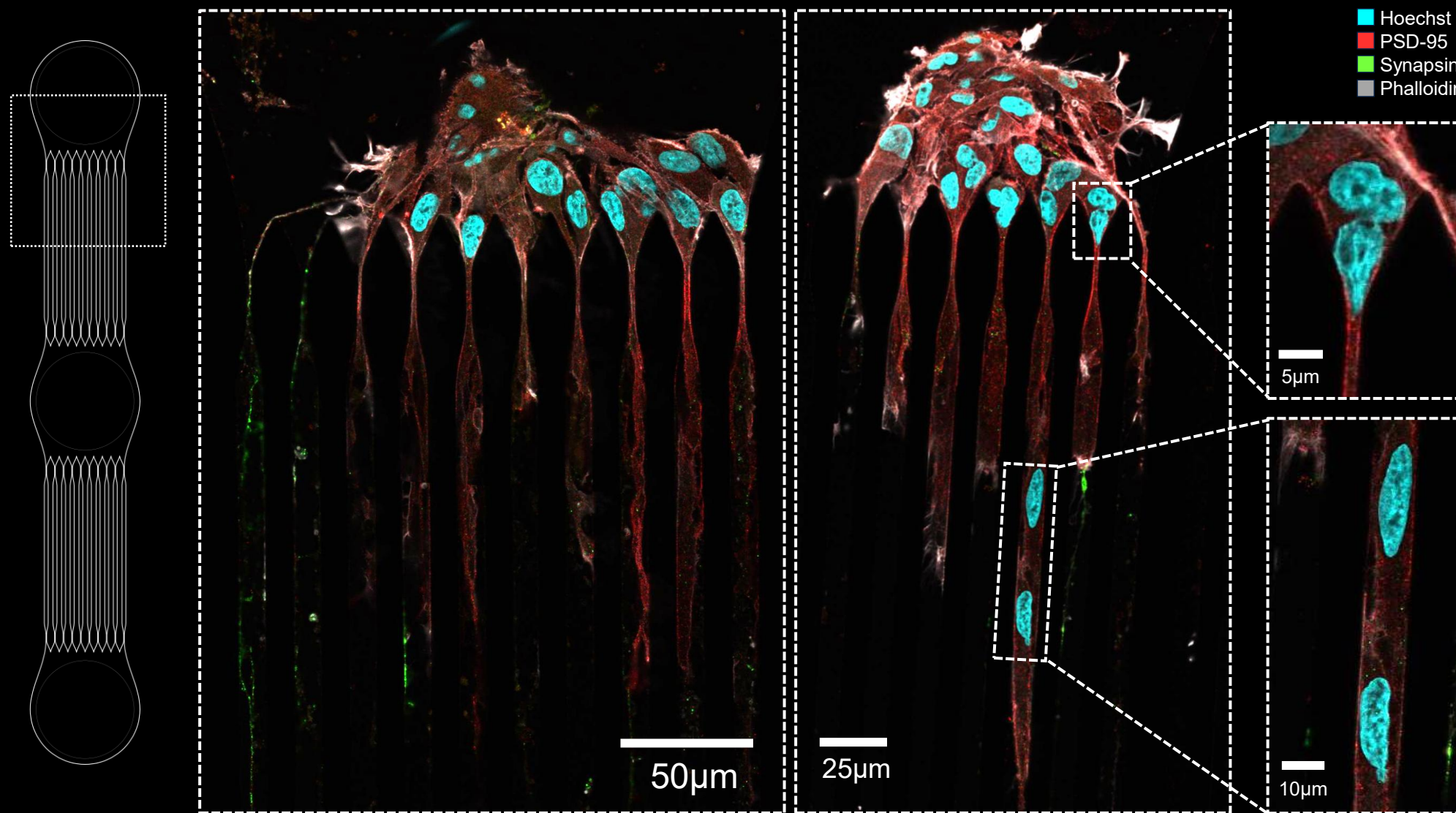


0-10 hrs

- Cell tracker (Glioblastoma)
- Cell tracker (Neurons)



Glioblastoma cells extend cell processes into the microchannels

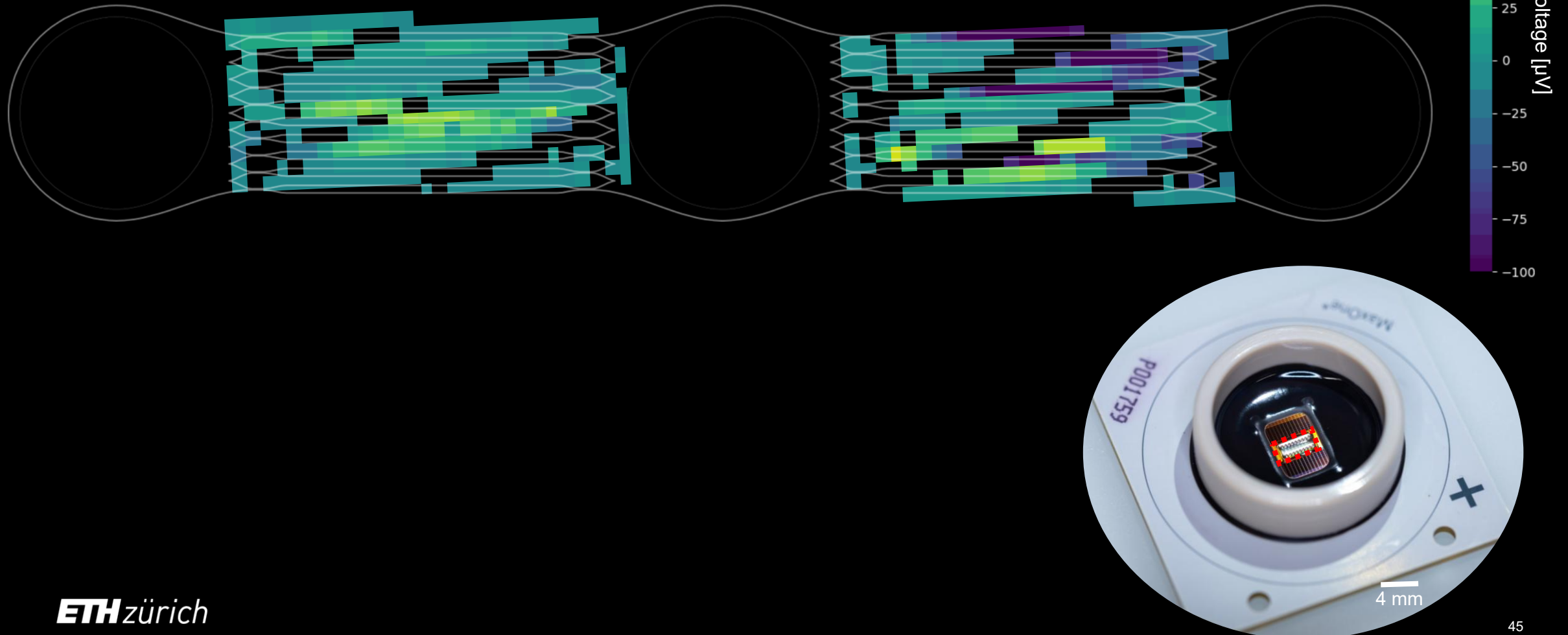


We can record neuronal activity with high spatial and temporal resolution

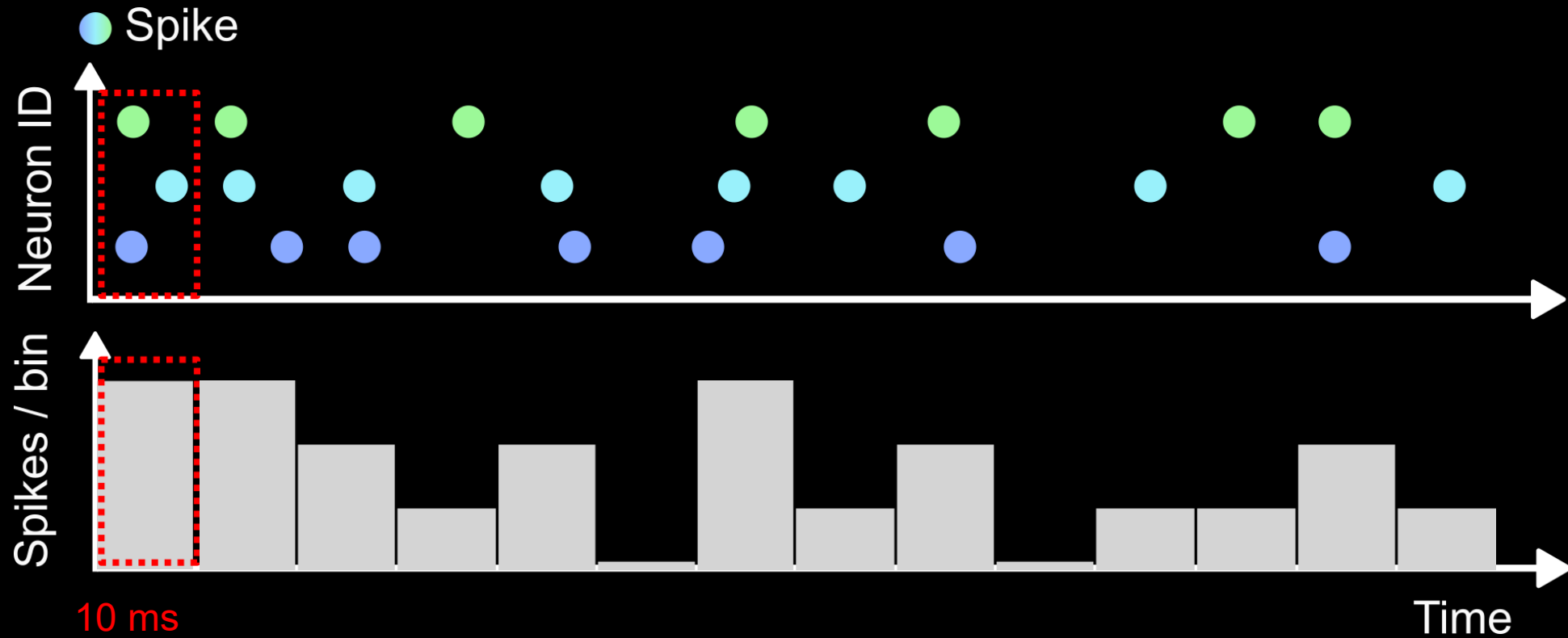
Disease modelling

Glioblastoma-Neuron Coculture (DIV 47)

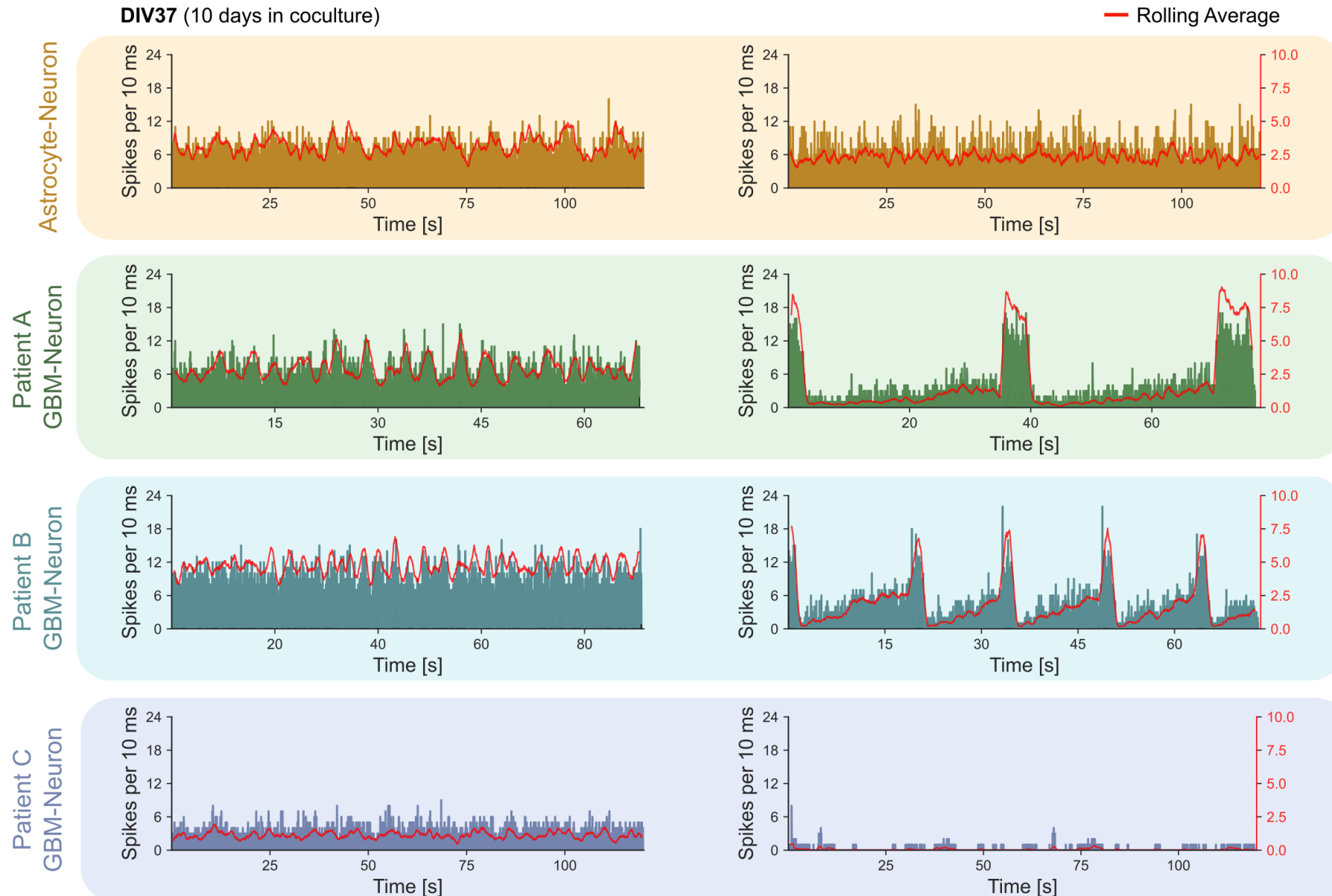
Playback speed: 0.5 ms / s



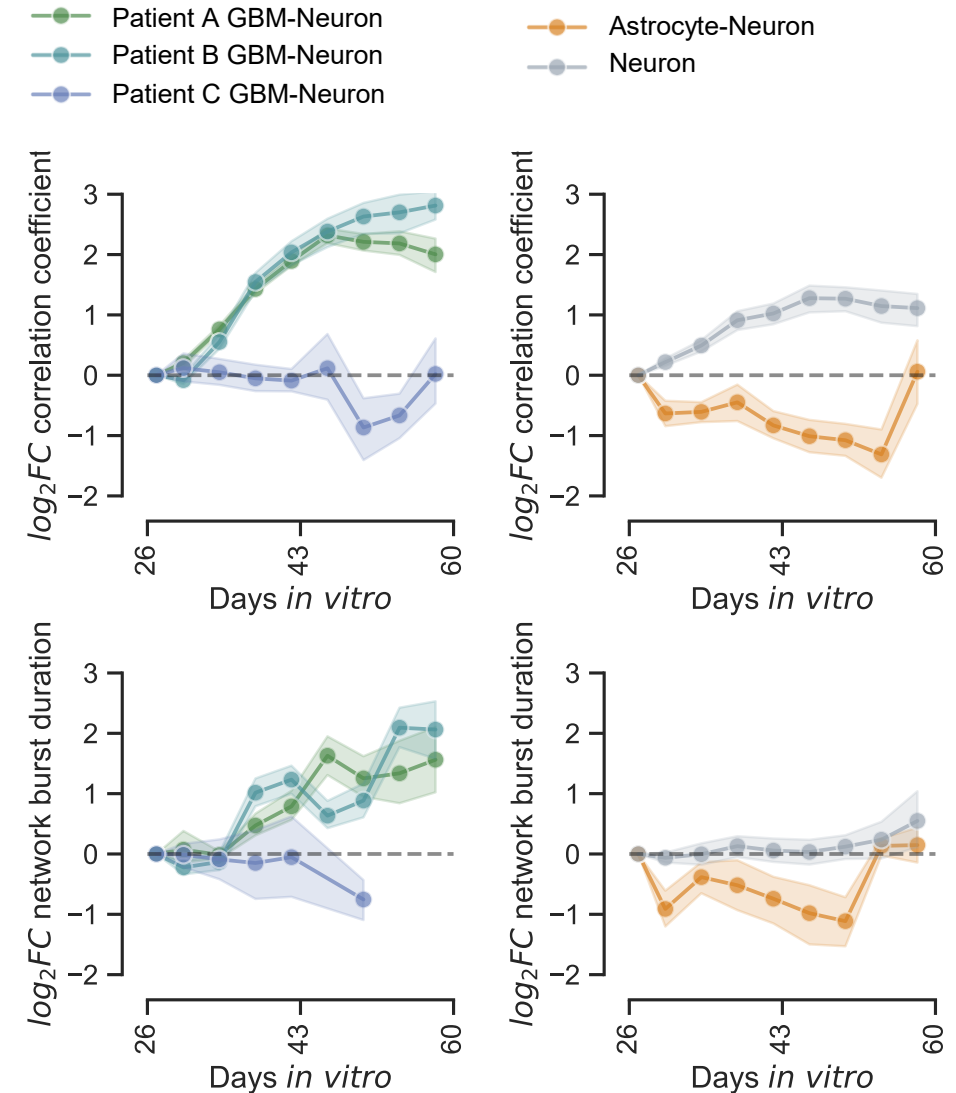
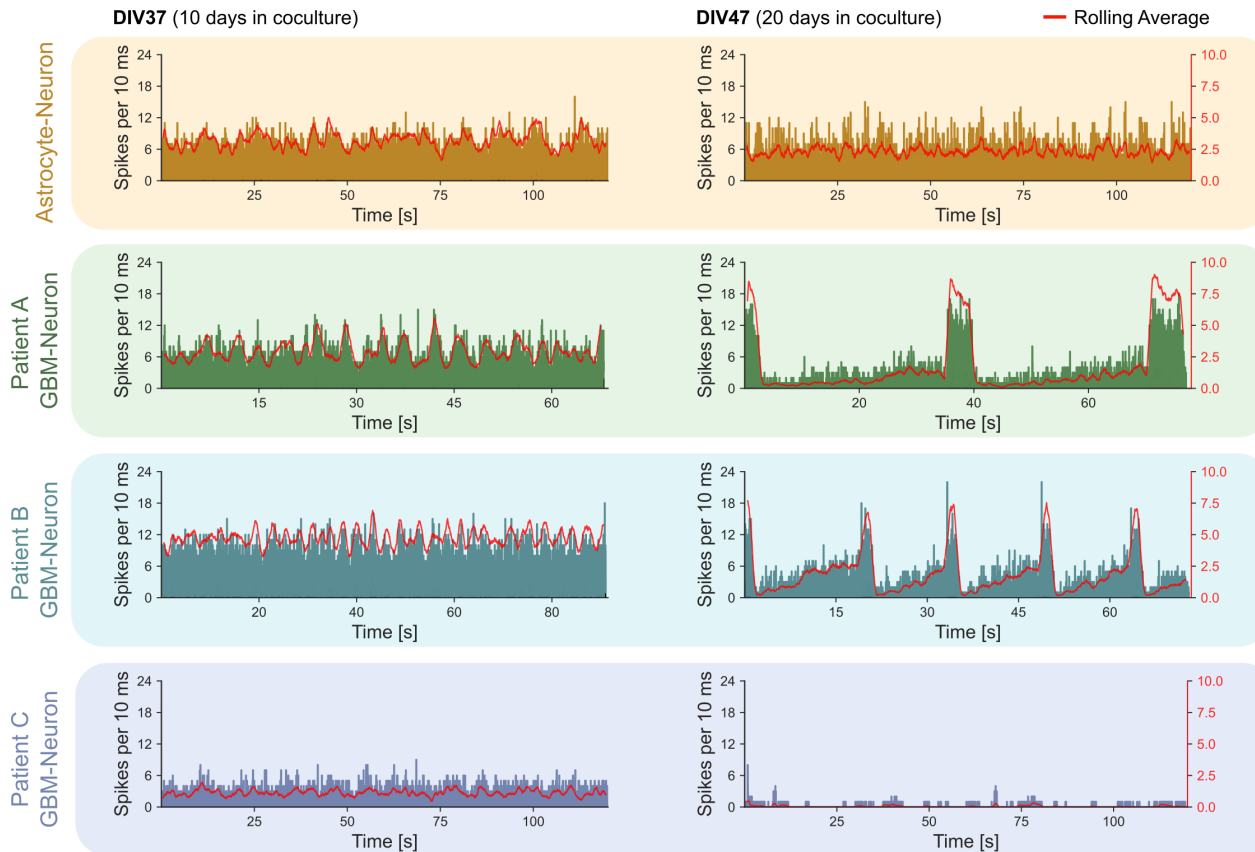
Firing rate histograms visualise the network activity over time

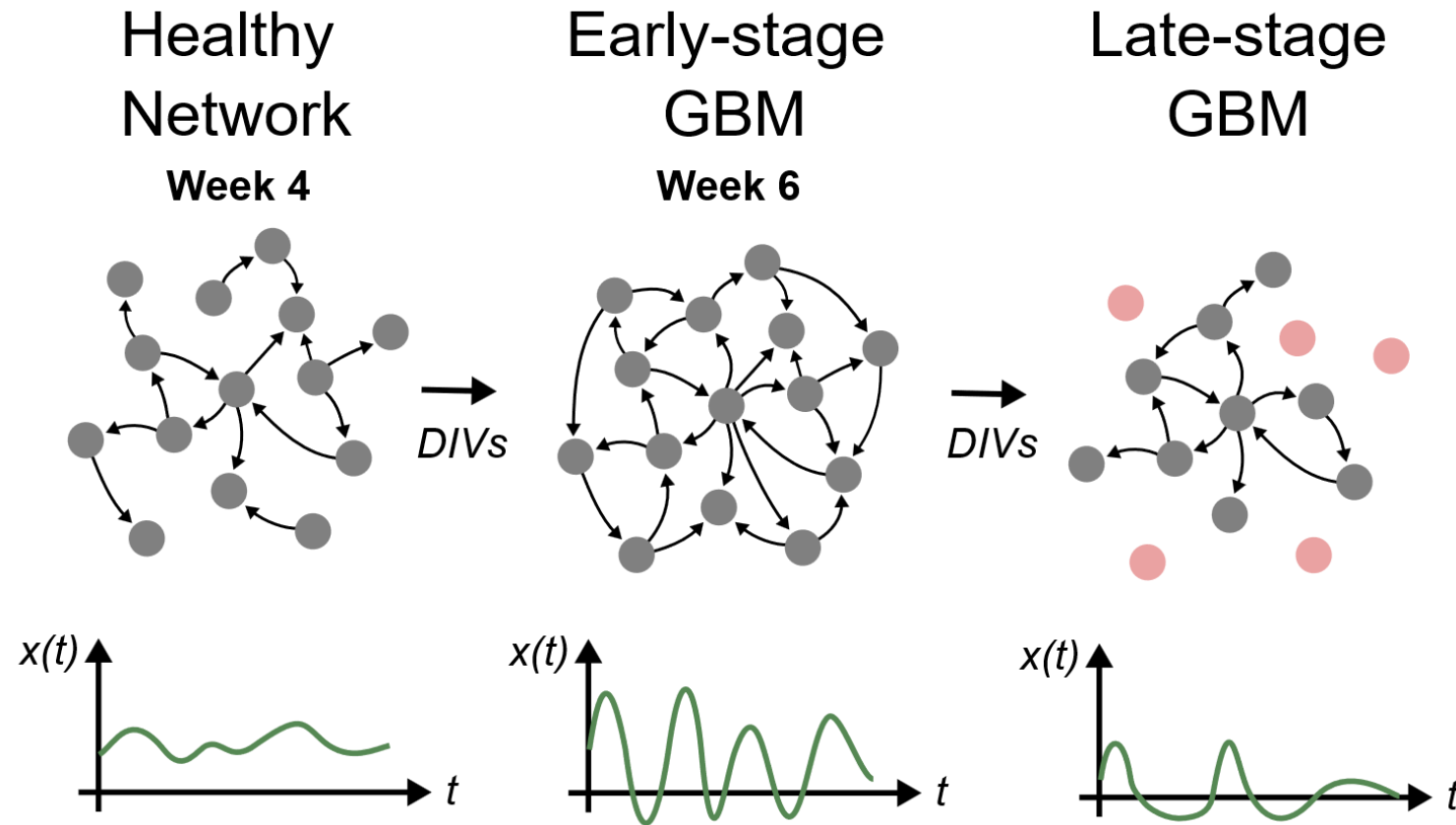


Characteristic large network bursts are recorded after 20 days

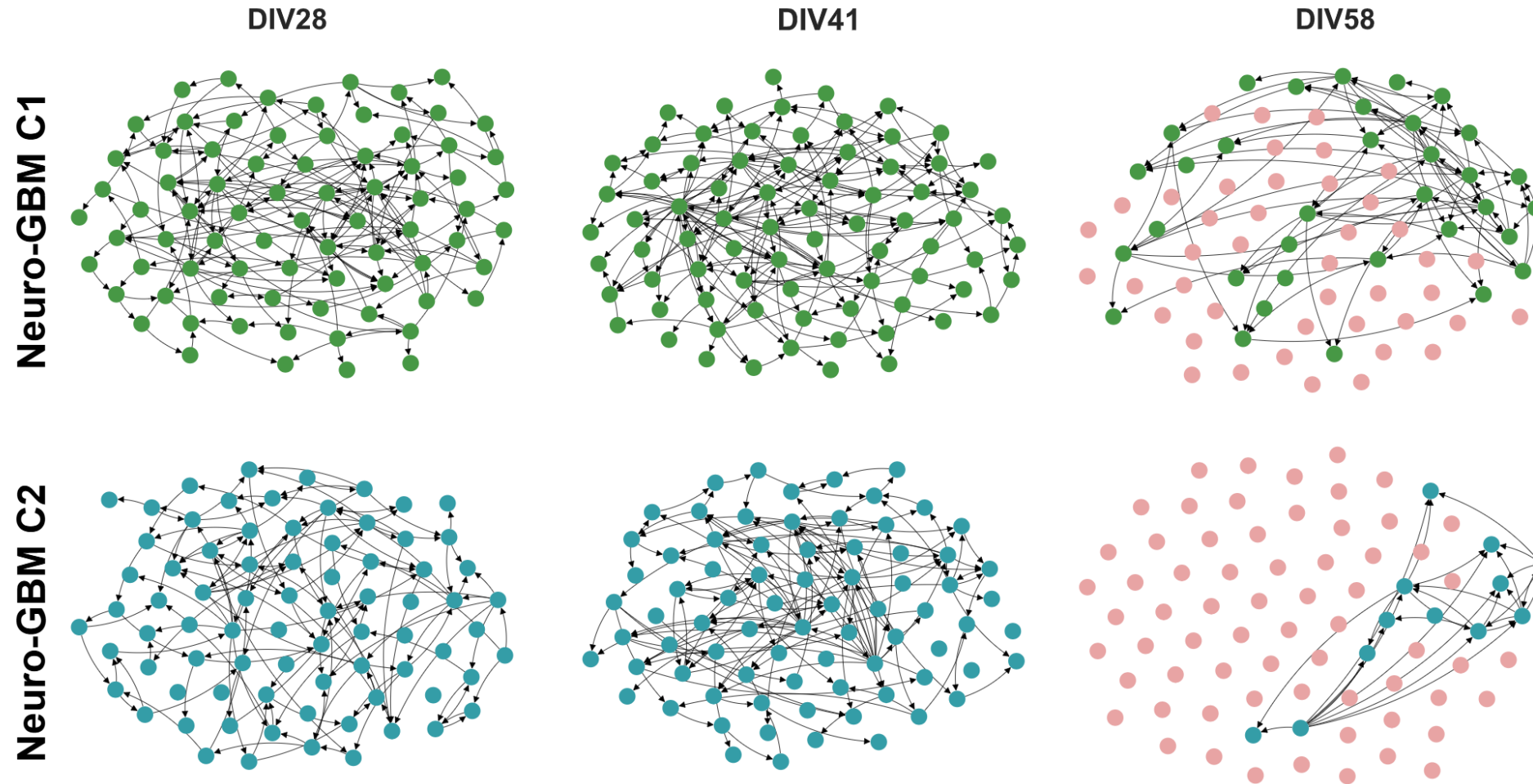


Characteristic large network bursts are recorded after 20 days

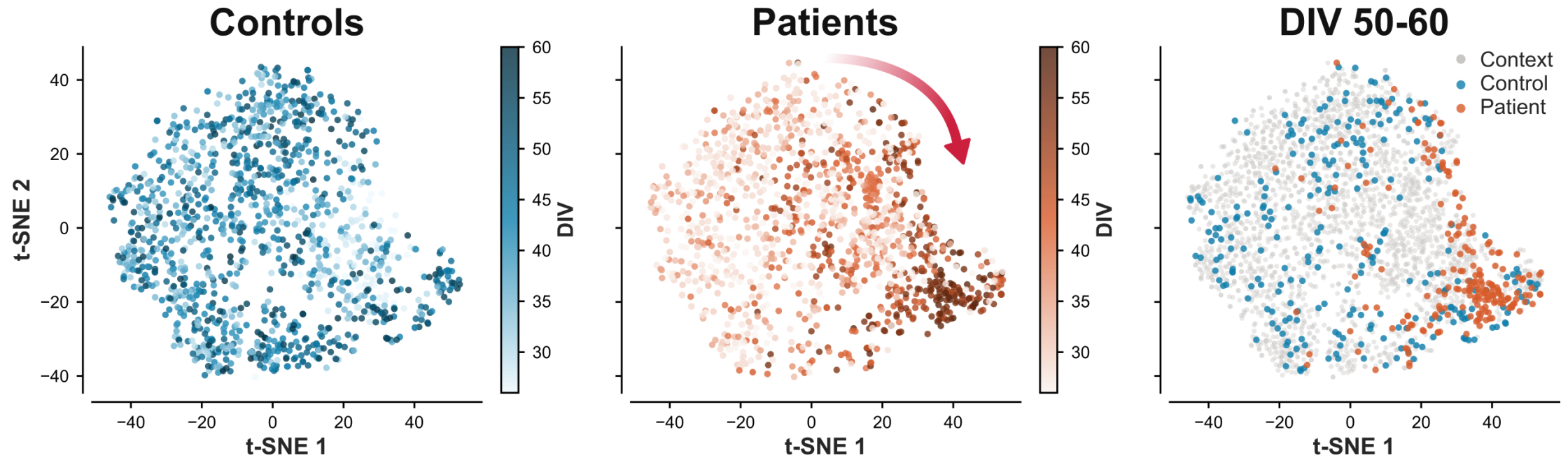




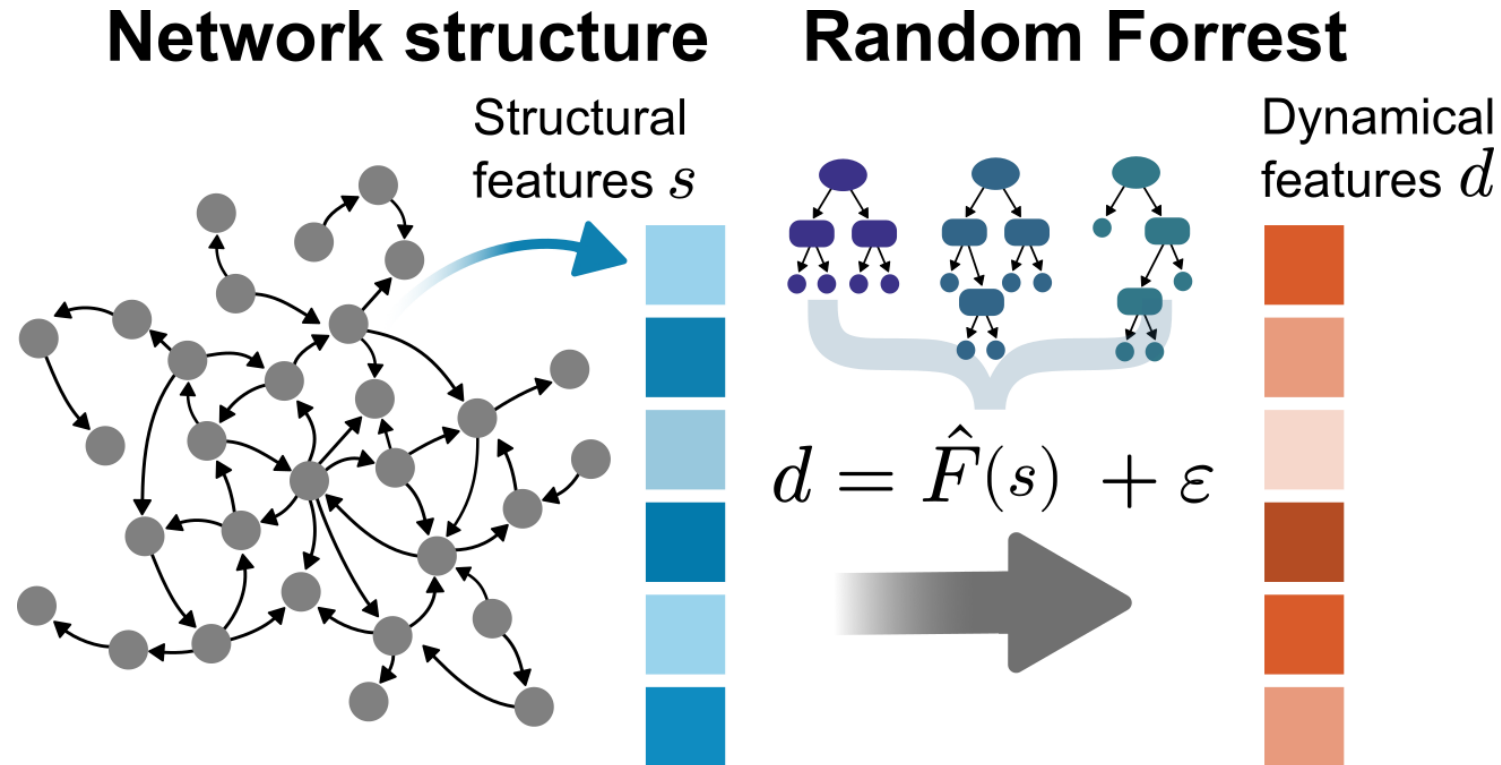
Deriving (effective) networks from neuronal spiking data



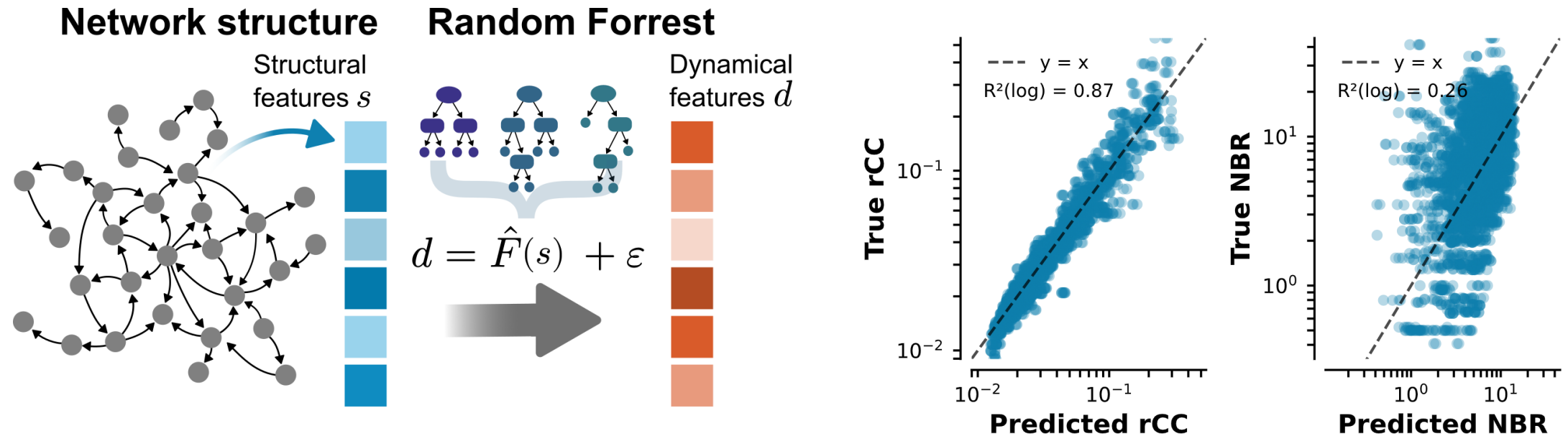
Spoiler: There should be a characteristic change in the network structure



Do these structural changes explain the network synchronization and network bursting?



Do these structural changes explain the network synchronization and network bursting?



THANK
YOU
BIORITC